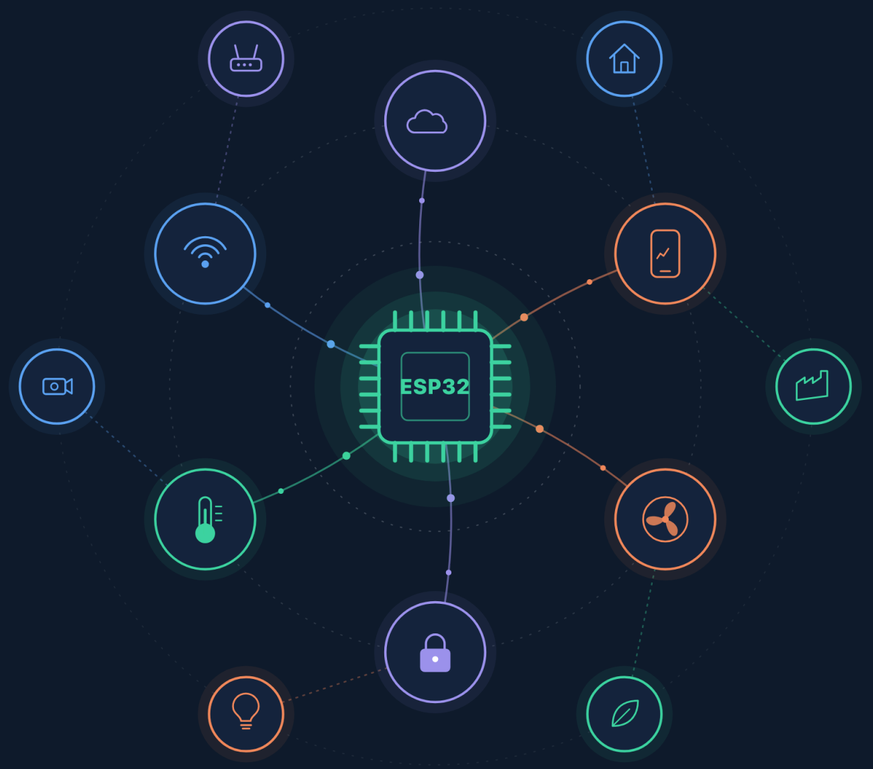
MODULE 1 · 45 MINUTES

What Is the Internet of Things?

What IoT is, what makes a device an IoT device, and how telcos already use it at tower sites and in data centres.



Mehdi Lotfinejad

Introduction to the Internet of Things

STORY ONE · THE TOWER SITE

A hilltop tower site has lost grid power. The generator is running. How much fuel is left?

The nearest engineer is hours away by road.

STORY TWO · OUR DATA CENTRE

2 a.m. In our data centre, one rack is getting hotter. Nobody is in the room.

Who finds out first: a sensor, an engineer, or a customer?

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
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You are here

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HOW WE WORK TODAY

One case, two lenses

One case

Start from a real problem: the tower site on generator, or the hot rack.

Technical lens

Asks **how**: which sensor, which network, edge or cloud?

Business lens

Asks **so what**: cost, uptime, risk, payback.

At every step today, we ask **both** questions.

BY THE END OF THIS MODULE

You will be able to...

01 **Explain** what IoT is, with four words: sense, send, decide, act

02 **Spot** IoT at a tower site and in our data centres

03 **List** the characteristics that make something an IoT device

04 **Name** five ways telcos use IoT, and why each one matters

05 **Ask** two questions about any IoT idea: how, and so what?

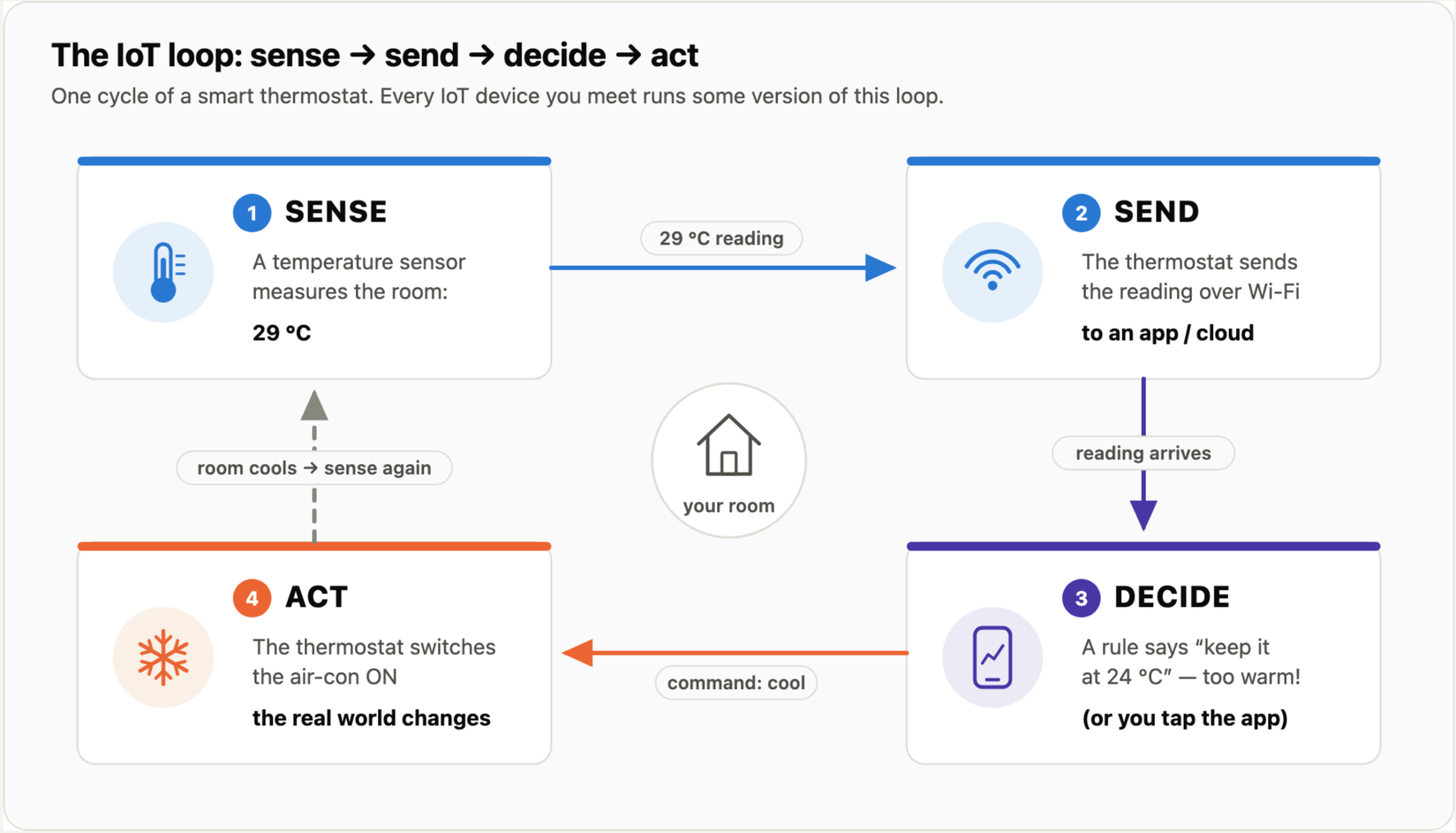
A SIMPLE DEFINITION

The Internet of Things is the network of physical objects that sense the world, share what they sense, and act on it.

A “thing” can be almost anything: a fuel tank, a shelter door, a water pump, a rack sensor, or a cow with a smart collar.

THE IOT LOOP

Sense, send, decide, act



PAUSE AND CHECK

At a tower site, a sensor measures the fuel in the generator’s tank and reports it to the NOC. When fuel is low, someone sends a truck. Can you find the four steps?

Sense → send → decide → act. Talk to your neighbour for one minute.

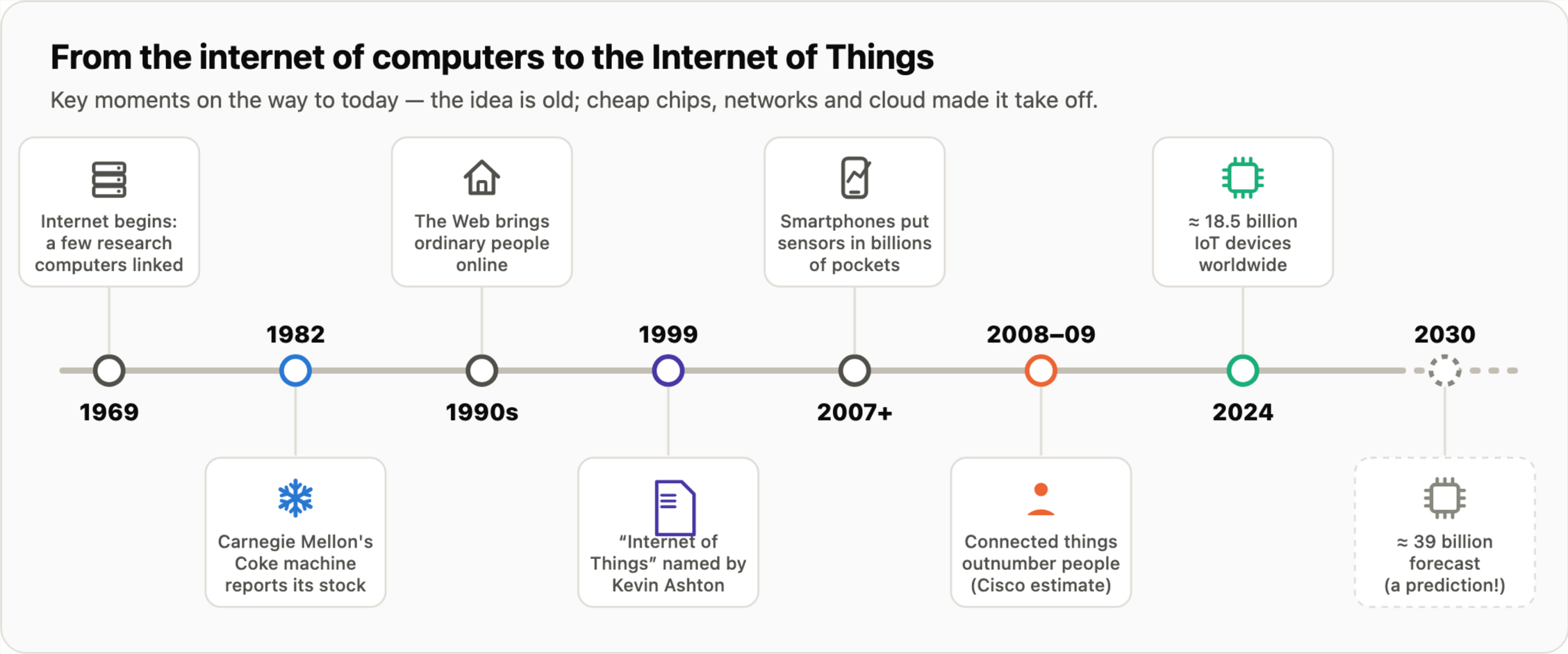
Part A

How we got here

From an internet of computers to an Internet of Things.

SIXTY YEARS IN ONE LINE

From a few computers to billions of things



WHAT MAKES A DEVICE AN IOT DEVICE?

Eight shared characteristics

Characteristic	What it means	Example
Sensing	Measures something in the physical world	A fuel sensor measures the tank
Connectivity	Sends and receives data over a network	The site reports to the NOC
Unique identity	Each device can be told apart	Each site and sensor has its own ID
Automation	Processes data and acts by itself	Extra cooling switches on at 28 °C
Ability to act	Changes the physical world	A relay starts the generator
Scale	Often huge numbers of devices	Many sites, watched from one NOC
Limited resources	Little power, memory and processing	A small sensor on a battery
Variety	Many makers, still working together	Generators and batteries from many brands

If you remember only two: it **senses or acts** on the physical world, and it is **connected**.

Part B

Why now — and why here?

The Coke machine was 1982. Why is IoT booming now, and why does it matter for a telco?

FOUR THINGS GOT CHEAP AT ONCE

Why IoT is growing so fast

Sensors

Smartphones created a huge market for tiny sensors, so prices fell for everyone.

Chips

A microcontroller with Wi-Fi built in, like the ESP32, now costs very little.

Connectivity

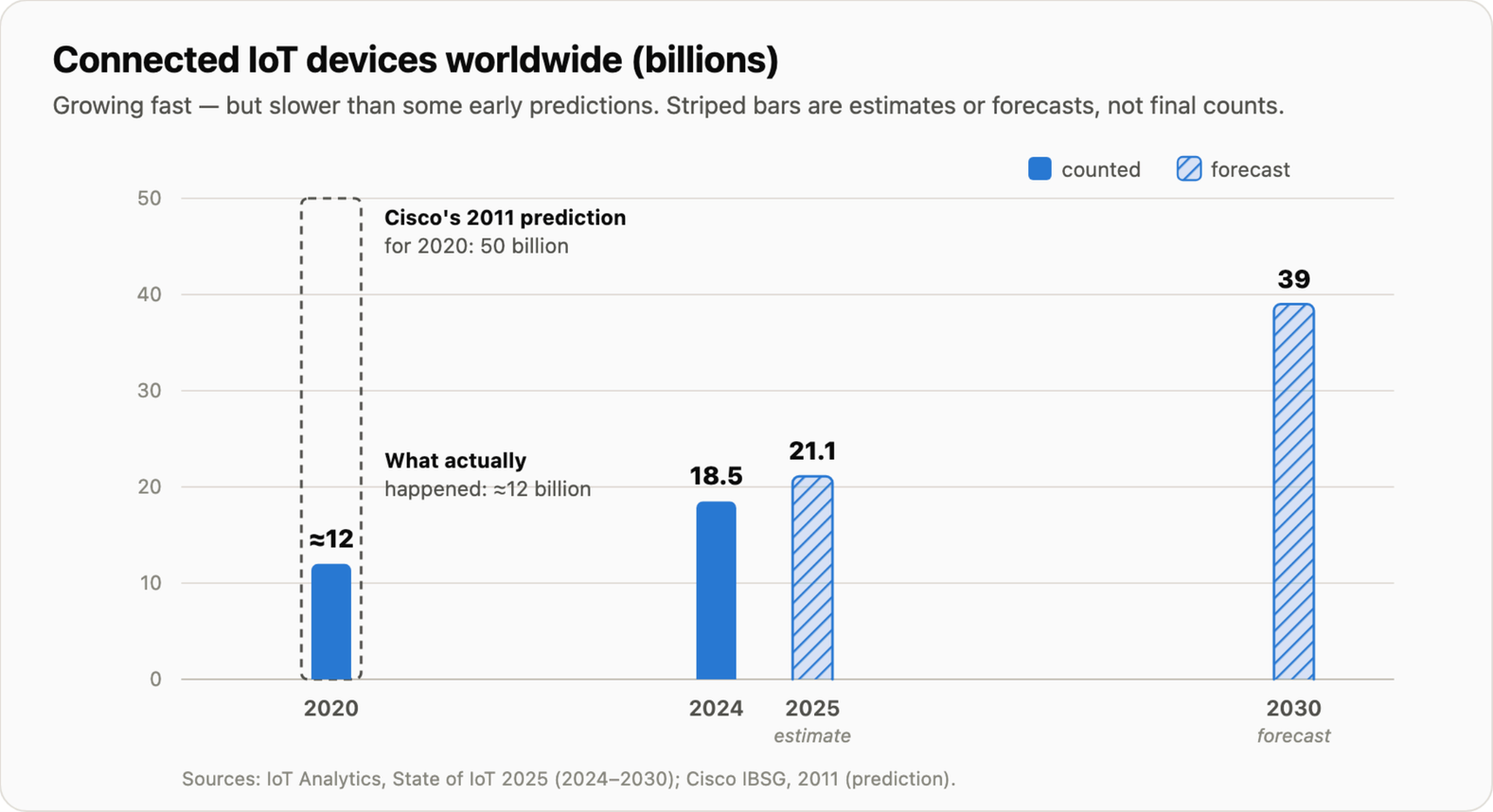
Mobile networks are almost everywhere, even at remote tower sites.

The cloud

Companies rent storage and computing by the minute, instead of buying servers.

HOW BIG IS IT?

Billions of connected devices



18.5 billion IoT devices in 2024

39 billion forecast for 2030

2020: forecast **50 billion**, real **12 billion**

Source: IoT Analytics. Dashed outline: Cisco’s 2011 forecast for 2020.

WHY THIS MATTERS HERE

Why this matters for a telco

15–20%

of a mobile operator's **operating costs** go on energy (GSMA)

1 bn

low-power cellular IoT **connections** worldwide, end of 2025

2,055

MW of data-centre capacity in Malaysia by end-2026: double

More sites, more energy, more to watch: things that **report on themselves**.

Source: GSMA Intelligence (2025); GSMA; JLL, via The Edge Malaysia (April 2026).

Part C

Where do telcos use IoT?

Five ways telcos use IoT — and our two cases, up close.

WHERE IOT IS USED

Five ways telcos use IoT

Smart tower monitoring

The NOC sees the health of every site, live.

Remote site monitoring

Fuel, battery and door, without a site visit.

Predictive maintenance

Fix batteries and generators before they fail.

Energy management

Less diesel: run the generator only when needed.

Asset tracking

Where are our generators, spares and vans?

Each one comes back **later today**.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

CASE ONE · THE TOWER SITE

IoT at the tower site

- **Sense:** fuel level, battery voltage, shelter door
- **Send:** site RMS → the NOC, by mobile network
- **Decide:** fuel falling while the generator is off?
- **Act:** send fuel or security in time
- **So what?** Fewer trips, less theft, sites on air



Photo: BalticServers.com, CC BY-SA 3.0, via Wikimedia Commons

CASE TWO · OUR DATA CENTRES

IoT in our data centres

- **Sense:** temperature, humidity, leaks, doors
- **Send:** to DCIM and the operations centre
- **Decide:** over 27 °C? (ASHRAE: 18–27 °C)
- **Act:** extra cooling on, engineer alerted
- **So what?** Cooling uses 7% to over 30% of a site’s electricity (IEA)

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ **Myth** *IoT means controlling things from your phone.*	✓ **Fact** Most IoT data never reaches a phone. Other computers use it.
✗ **Myth** *IoT is just gadgets for the home.*	✓ **Fact** Tower sites, data centres, factories and cities are a huge share.
✗ **Myth** *An IoT device must be powerful.*	✓ **Fact** Most are tiny, cheap and low-power — on purpose.
✗ **Myth** *IoT is a single technology.*	✓ **Fact** It combines sensors, chips, networks, cloud, data and security.

MIXED TEAMS OF THREE

Three roles that rotate each lab

Builder

Hands on the keyboard: pastes the code, presses Play.

Analyst

Reads the guide aloud, checks the output, writes down numbers.

Decision owner

Answers the manager task: which alerts and actions matter?

Roles **rotate** each lab, so every manager builds at least once.

TEAM ACTIVITY · 5 MIN

Where is IoT already working in your site?

1. Form a team of three: mix technical and business people.
2. Choose your first roles: builder, analyst, decision owner.
3. Discuss: where is IoT already working at your sites today?
4. Pick one example and find the loop: sense → send → decide → act.
5. Decision owner: add one business question (cost, uptime or risk).

IN TELCO OPERATIONS

What IoT changes at our sites

Today	With IoT	The benefit
Engineers drive out to check each site	The site reports fuel, battery and door	Fewer site visits
Fuel trucks run on a fixed schedule	A fuel sensor tells the NOC the level	Fewer trips; no site runs dry
Fuel theft is found days later	Alarm: fuel drops, generator off	Less theft, faster response
The NOC sees an alarm, not the cause	The site sends the readings with the alarm	Faster, better-aimed repairs

Every module asks this question: what changes in our **daily work**?

KEY TAKEAWAYS

What to remember

- IoT = physical objects that **sense, share and act** on data
- Every IoT system follows the loop: **sense → send → decide → act**
- Telcos use IoT for **towers, sites, maintenance, energy and assets**
- Our two cases all day: **the tower site**, then **our data centres**
- Two lenses: **how** does it work? **So what** for cost, uptime and risk?
- Mixed teams of three; **roles rotate** each lab

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 One shelter thermometer shows the temperature on its own screen. Another sends it to the NOC. Which one is IoT — and why?

Q2 Apply the four steps to the generator’s fuel tank at a tower site.

Q3 Name two of the five ways telcos use IoT, and one benefit of each.

Q4 Which lens? (a) Which sensor? (b) What does an outage cost? (c) Edge or cloud?

UP NEXT · MODULE 2

How an IoT System Is Built

We open the “black box”: the five layers of every IoT system, and the journey of one reading from a shelter sensor to the NOC.

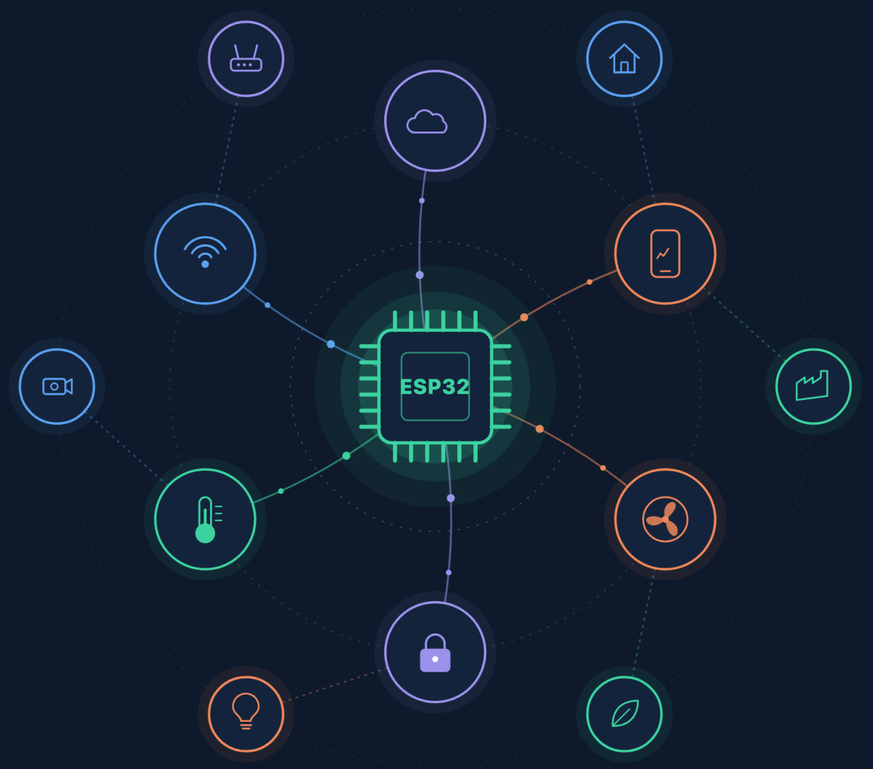
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 2 · 45 MINUTES

How an IoT System Is Built

The five layers inside every IoT system, from a tower site to a data centre.



Mehdi Lotfinejad

Introduction to the Internet of Things

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BY THE END OF THIS MODULE

You will be able to...

01 **Name** the five layers of an IoT system and what each one does

02 **Map** the five layers onto a tower site and a telco data centre

03 **Explain** the difference between a sensor and an actuator

04 **Compare** a microcontroller with a microprocessor

05 **Weigh** build, buy or telco-managed, and selling it as a product

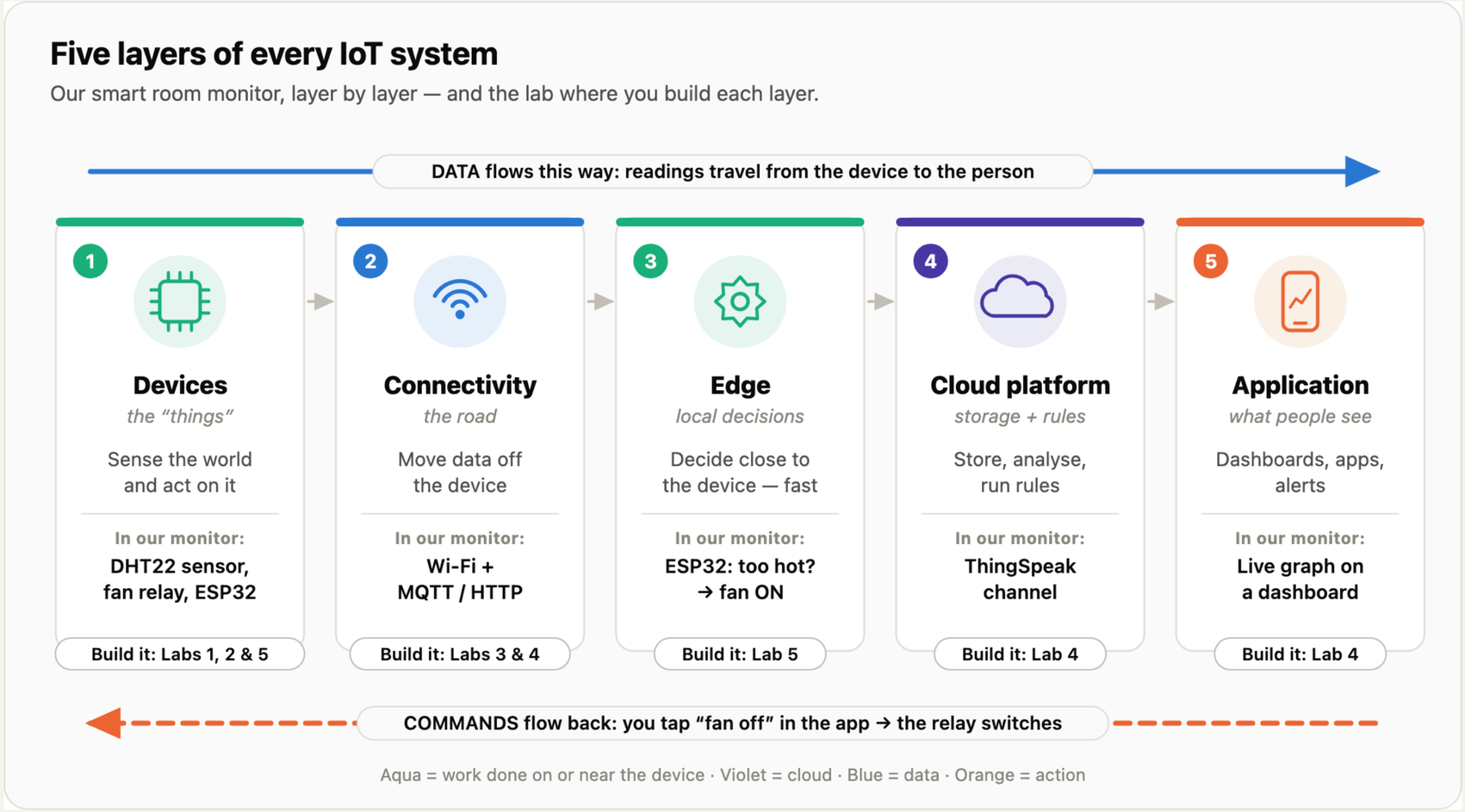
Part A

The five layers

One Smart Site Monitor, broken down layer by layer.

THE BIG PICTURE

Five layers of every IoT system



LAYER 1 · DEVICES

Three ingredients in every device

Sensors

Measure something, like shelter temperature or fuel level.

Actuators

Do something, like switching on extra cooling.

A small computer

Usually a microcontroller that runs the program.

In the labs, your device is a simulated **ESP32** board.

LAYER 2 · CONNECTIVITY

The road the data travels on



The link

How the signal travels: a mobile network, a cable or Wi-Fi.



The language

What the device speaks, like MQTT or HTTP.



A gateway

A middleman box that serves many nearby devices.

At a tower site, the site’s monitoring box is the gateway. At home, your Wi-Fi router is one.

LAYER 3 · EDGE COMPUTING

Edge computing means doing some of the thinking close to where the data is collected.

On the device itself, or on a gateway nearby — not far away in the cloud. It is faster, and it keeps working when the connection doesn’t.

LAYER 4 · CLOUD PLATFORM

What a cloud platform does

Stores

Keeps old readings, so you see history.

Processes

Works things out, like a daily average.

Runs rules

“Above 28 °C? Send me a message.”

Manages devices

Knows which devices exist and may connect.

In Lab 4 you use a free cloud platform, **ThingSpeak**. Module 5 covers this layer.



Photo: Daoducquan, CC BY-SA 4.0, via Wikimedia Commons

LAYER 5 · APPLICATION

What people see and use

- **Dashboards** in the NOC, with live graphs
- **Alerts** by app, SMS or email
- **Reports** for managers
- **Work orders** that send a field engineer
- **Other systems**, like billing

PAUSE AND CHECK

Our Smart Site Monitor switches on extra cooling in the shelter by itself, without asking the cloud. Which layer is making that decision?

Devices · Connectivity · Edge · Cloud platform · Application

ONE MAP, TWO CASES

The five layers at a tower and a data centre

Layer	Tower site	Telco data centre
1 Devices	Shelter heat, fuel, battery, door	Rack heat, humidity, leak, door
2 Connectivity	Cellular link (mobile network)	Cable or Wi-Fi
3 Edge	Site monitoring box (RMS)	Gateway, BMS or DCIM on site
4 Platform	Monitoring platform in the NOC	Monitoring platform
5 Application	NOC alarms, field engineer sent	Operations dashboard and alerts

RMS remote monitoring system · **NOC** network operations centre · **BMS** building management system · **DCIM** data-centre infrastructure management

Part B

Inside the device

Senses, muscles and a brain.

THE DEVICE’S SENSES AND MUSCLES

Sensors and actuators

	Sensor (input)	Actuator (output)
Job	Measures the world	Changes the world
Like your…	Senses	Muscles
Examples	Temperature, fuel-level, door and smoke sensors	LED, buzzer, motor, relay (an electric switch)
In our monitor	Shelter temperature and humidity sensor	Relay that switches extra cooling

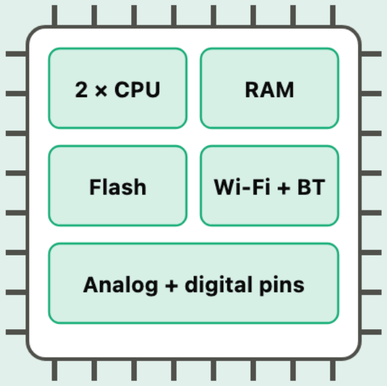
Ask: **which way is the information going?** In from the world = sensor. Out into the world = actuator.

One chip, one job vs a small computer

One chip, one job vs a small computer

A microcontroller packs everything onto one chip. A microprocessor needs a board of helpers and an OS.

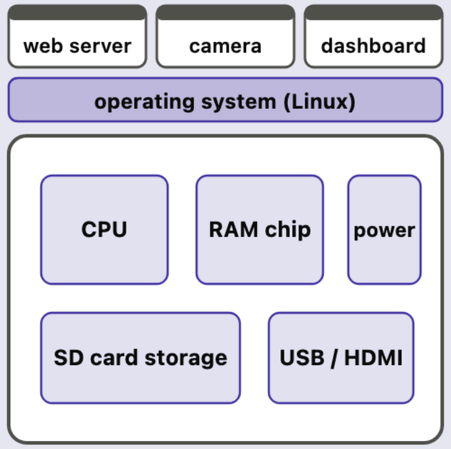
MICROCONTROLLER (e.g. ESP32)



everything on ONE chip

- **Runs ONE program,** from the moment power is on
- **Starts instantly** no booting
- **Deep sleep ≈ 10 µA** months–years on batteries
- **Low cost** cheap enough to put in anything

MICROPROCESSOR BOARD (e.g. Raspberry Pi)



several chips on a board

- **Runs an OS +** many programs at once
- **Boots in seconds** like a small PC
- **Draws watts** usually mains-powered
- **Great for video, AI** and heavy software

SIDE BY SIDE

Microcontroller or microprocessor?

	Microcontroller (MCU)	Microprocessor (MPU)
Like a…	Kitchen timer: one job, done reliably	Laptop: many jobs, more looking after
Runs	One program, straight away	An operating system and many programs
Power use	Very low: months or years on batteries	Higher: usually mains power
Start-up	Almost instant	Seconds to a minute (it boots)
Cost	A few US dollars	Tens of US dollars and up
Good at	Reading sensors, switching things	Video, heavy data, a local dashboard
Examples	ESP32, Arduino boards, STM32	Raspberry Pi, the chip in your phone

Reads sensors and flips switches? **Microcontroller.** Video or lots of software? **Microprocessor.**

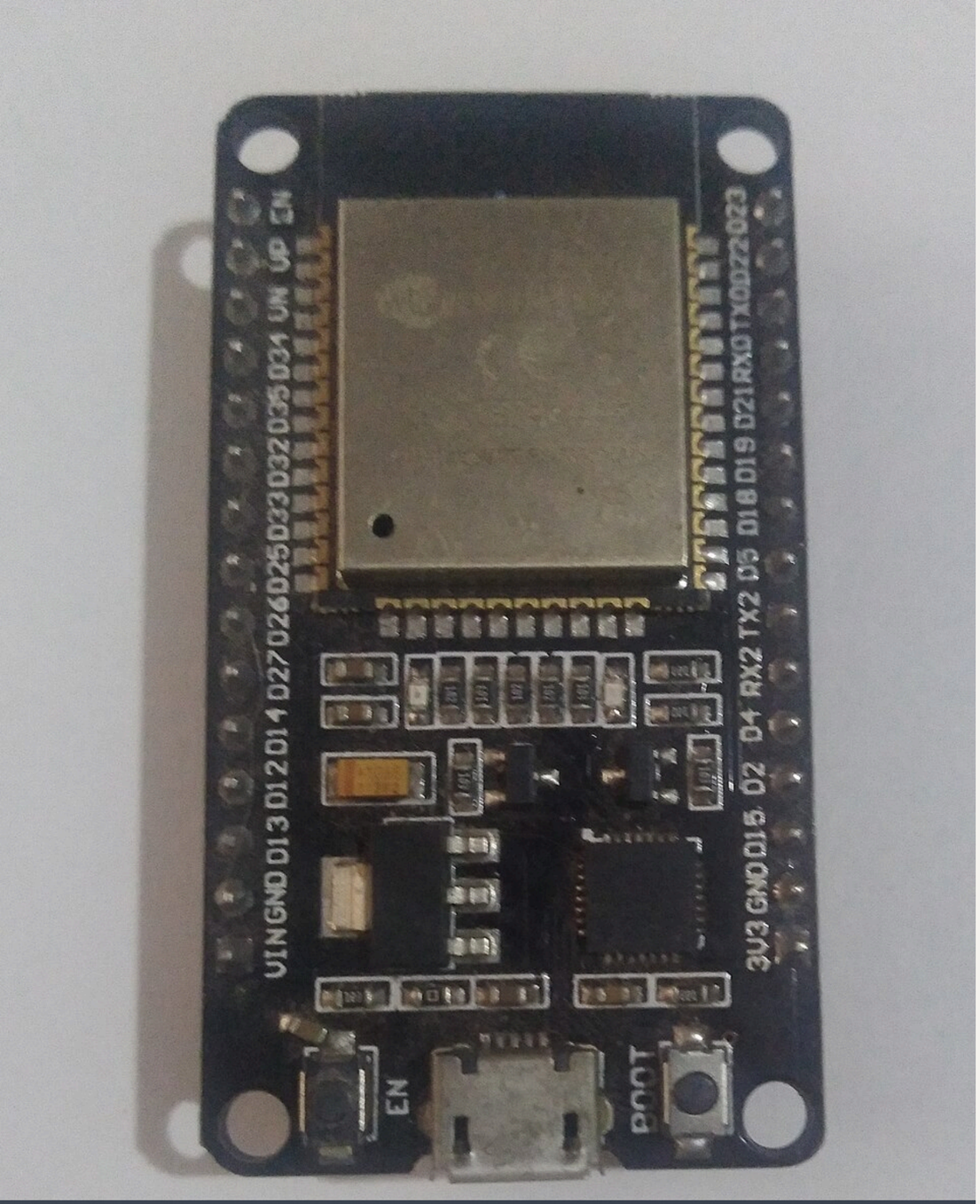


Photo: Edwiyanto, CC BY-SA 4.0, via Wikimedia Commons

THE DEVICE YOU'LL USE

Meet the ESP32

- The brain of our **Smart Site Monitor**
- **Wi-Fi and Bluetooth** built into the chip
- **Cheap**, and uses little power when resting
- **Lots of pins** for sensors and actuators

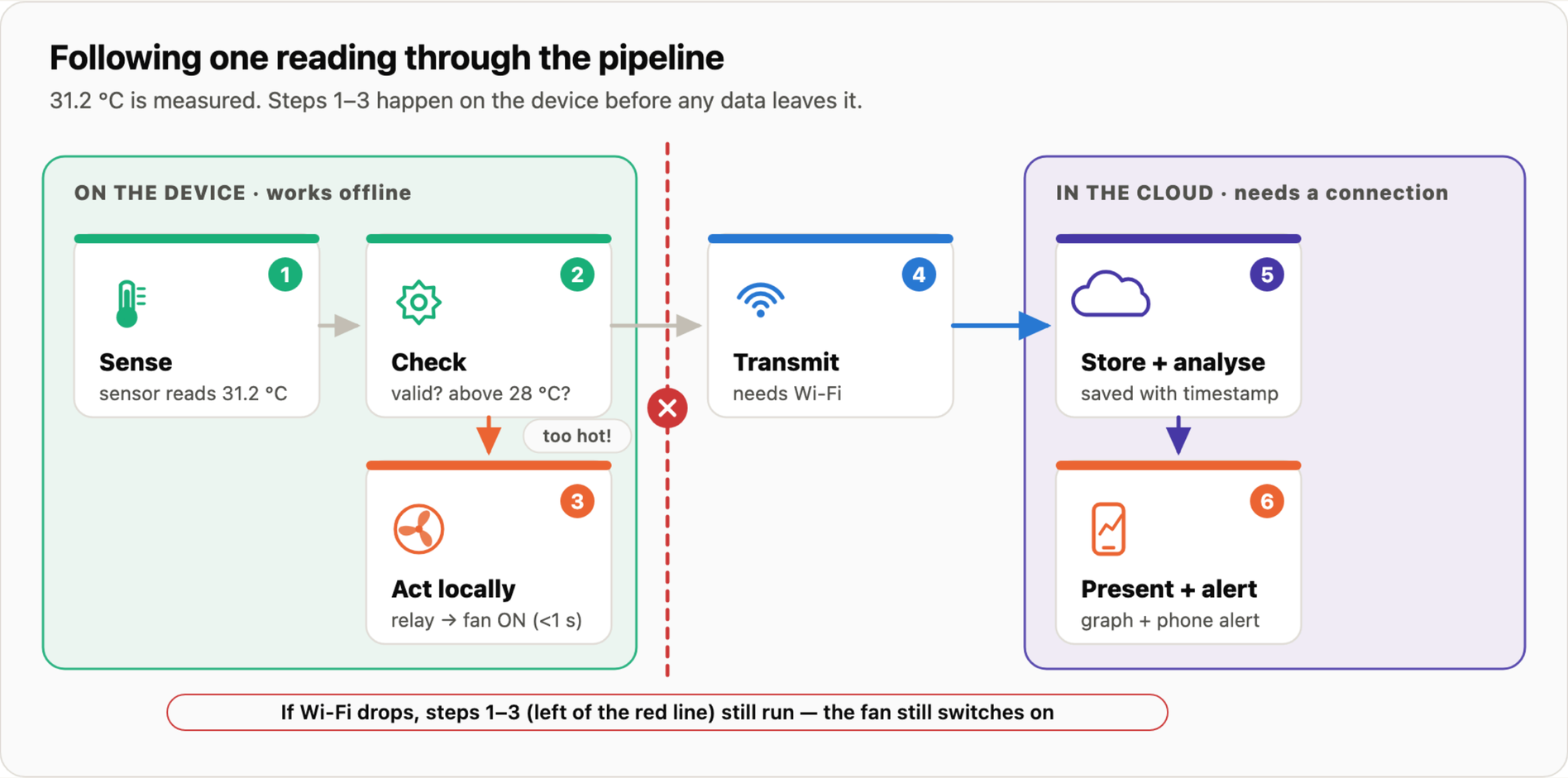
Part C

One reading, and who builds it

From a sensor, through the layers, to a person — and three ways to get there.

THE DATA PIPELINE

One reading’s journey



PAUSE AND CHECK

In the pipeline, which steps would still work if the network link went down?

1 Sense · 2 Check · 3 Act locally · 4 Transmit · 5 Store · 6 Present. Talk to your neighbour for one minute.

BUSINESS LENS · WHO BUILDS IT?

Build, buy or telco-managed



Build in-house

Cost: parts and staff time. **Speed:** slowest. **Control:** full.



Buy from a vendor

Cost: price plus licences. **Speed:** fast. **Control:** the vendor’s.



Telco-managed

Cost: a monthly fee. **Speed:** fast. **For a telco:** also a product to sell.

Telcos don’t only use IoT, they sell it: **Maxis** sells NB-IoT, 4G M2M and 5G IoT plans (Maxis Business website).

COMMON MISCONCEPTIONS

Three ideas that trip people up

✗ **Myth** *An IoT device must connect directly to the internet.*	✓ **Fact** Many talk only to a nearby gateway, which reaches the internet.
✗ **Myth** *The cloud does all the thinking.*	✓ **Fact** More and more decisions happen at the edge: faster, and offline too.
✗ **Myth** *A Raspberry Pi and an Arduino are the same thing.*	✓ **Fact** Different families: an OS on a microprocessor vs one program.

TEAM ACTIVITY · 5 MIN

Map a tower site or a telco data centre

1. Default: a tower site. Or choose a telco data centre.
2. Draw five boxes: devices, connectivity, edge, cloud, application.
3. Put one real part of the site into each box.
4. Which decision must happen at the edge? Why?
5. Decision owner: build, buy or telco-managed?

IN TELCO OPERATIONS

Smart tower monitoring, day to day

	Today	With IoT	The benefit
NOC alarm handling	Site off air; the NOC can’t see why	Sensors show why: power, heat, door	Faster fault repair
Field visits	Drive out, just to look	Check the site on the NOC screen	Fewer site visits
Site energy	Levels checked only on visits	Readings arrive in the NOC all day	Less diesel, less theft

Same five layers: sensors at the site, a cellular link, the RMS box, the NOC platform and alerts.

KEY TAKEAWAYS

What to remember

- Five layers: **devices, connectivity, edge, cloud, application**
- The same layers fit a **tower site** and a **data centre**
- Sensors bring information **in**; actuators push action **out**
- **Microcontrollers** do one job cheaply; **microprocessors** run an OS
- Urgent decisions stay at the **edge**, even when the link drops
- **Build, buy or telco-managed**; a telco can also **sell** it

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 List the five layers, in order from the device to the person.

Q2 At a tower site, is the shelter door contact a sensor or an actuator? And the relay that starts the generator?

Q3 A fuel sensor at a remote tower must run for years on a battery. Which brain: a microcontroller or a microprocessor?

Q4 A customer asks our telco to run their IoT. Name one strength and one risk.

UP NEXT · MODULE 3

Sensors, Actuators, and Reading Data

We zoom in on Layer 1: the sensors at a tower site and in a data centre. Then your first lab.

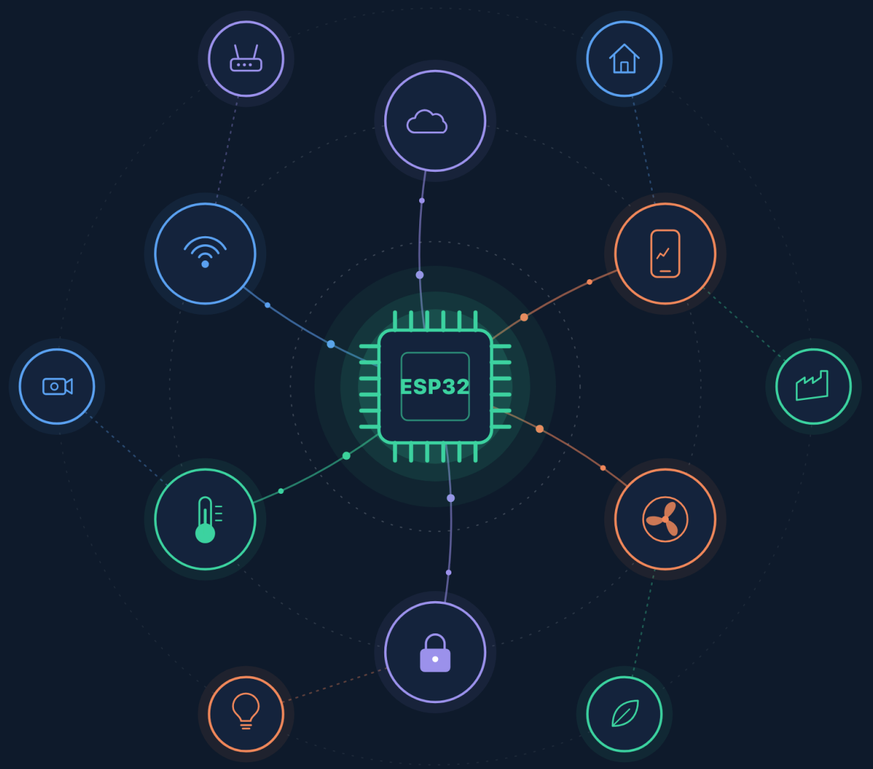
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 3 · 45 MINUTES

Sensors, Actuators, and Reading Data

How a device feels the air in a tower shelter, switches on extra cooling, and turns signals into numbers.



Mehdi Lotfinejad

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BY THE END OF THIS MODULE

You will be able to...

01 **Name** the sensors at a tower site and in a data hall

02 **Explain** how a relay lets a small device switch big equipment

03 **Tell** analog from digital, and how a signal becomes a number

04 **Build** a shelter sensor in the Wokwi simulator

05 **Weigh** the cost of a sensor against a site visit or an outage

THE BIG PICTURE

The journey of one reading

Real world

The air in the shelter is 24 °C.

Sensor

Turns heat into an electrical signal.

Pin

The microcontroller reads the signal.

Your code

Converts the signal into a number.

24.0 °C

Ready to show, or to send to the NOC.

Actions go the other way: code decides → a pin sends a signal → an actuator acts.

Part A

How devices feel the world

What we sense at a tower site and in a data hall.

REMOTE SITE MONITORING

Sensors at a remote tower site



Fuel level

How much diesel is left for the generator.



Battery voltage

Are the backup batteries charged and healthy?



Door contact

Is the shelter door open or closed?



Temp & humidity

The air in the equipment shelter. Too hot? Too damp?

Business lens: fuel that drops while the generator is **off** is suspicious. Possible theft.

OUR SECOND CASE

Sensors in our data centres and exchanges

Temp & humidity

The air going into each rack. Too hot? Too damp?

Water-leak cable

Lies on the floor. Says “wet” or “dry”.

Smoke detector

Early fire warning.

Power meter

A current clamp: how much power a rack uses.

The readings go to **DCIM** or the **BMS**: the software your operations team already uses.

THE BUSINESS LENS

A sensor versus an outage

No. 1

cause of serious outages: **power failures**

1 in 5

operators: last major outage cost **over US$1 million**

Ask: what does a sensor cost? A site visit? One missed warning? → **Canvas box 1**

Source: Uptime Institute, Annual Outage Analysis 2026

Part B

Actuators: how devices act

Light, sound, switching and movement.

FOUR COMMON ACTUATORS

From a signal to an action

LED

A status light: green for OK, red for a problem.

Buzzer

An alarm beep, to get attention.

Relay

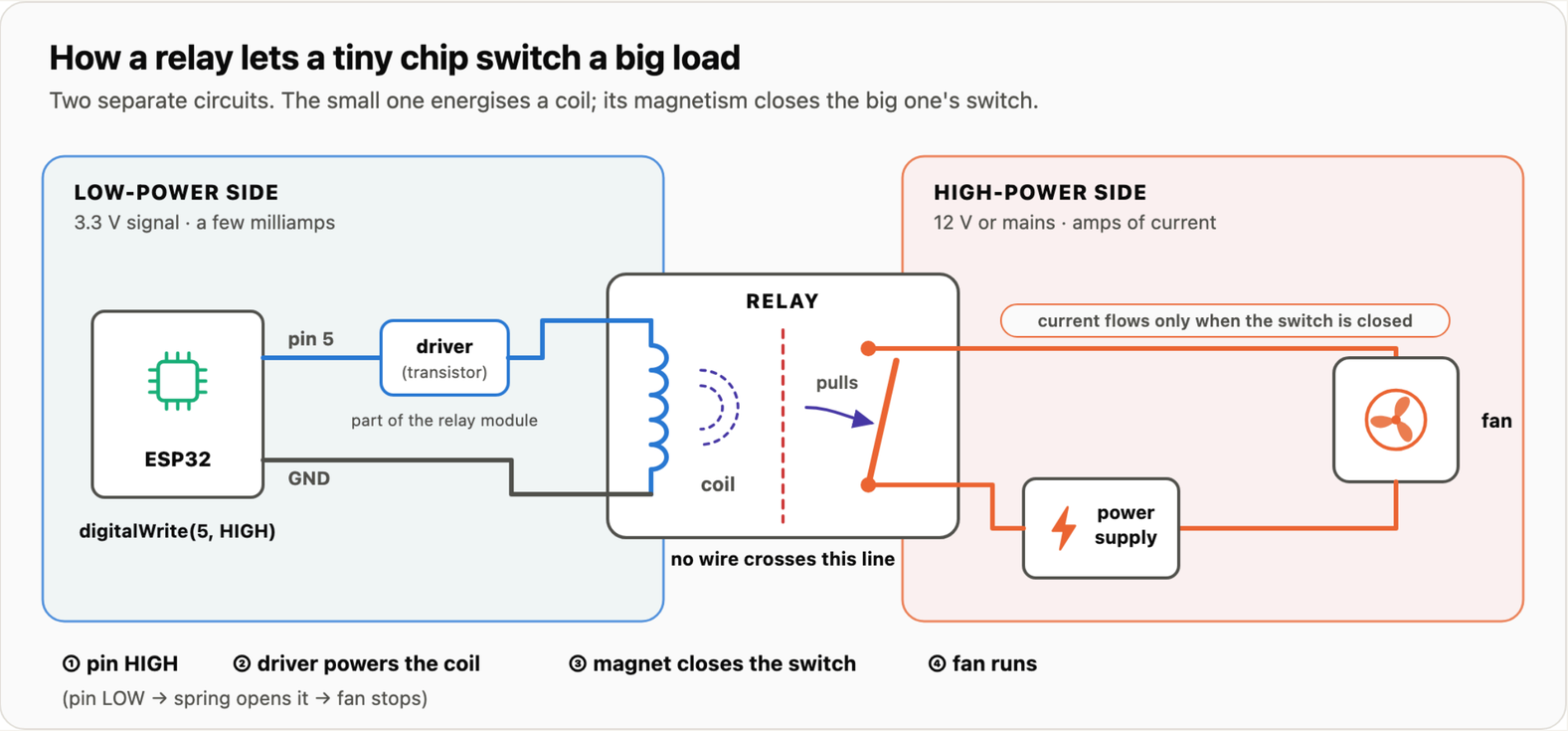
Switches big equipment, like extra cooling.

Motor

Movement, like opening a vent. Needs a driver.

HOW A RELAY WORKS

A small signal switches a big circuit



A pin gives only a **small** amount of power

The coil’s **magnetism** closes the switch

The two circuits stay **separate**

Part C

Analog vs digital signals

The most important technical idea today — and simpler than it sounds.

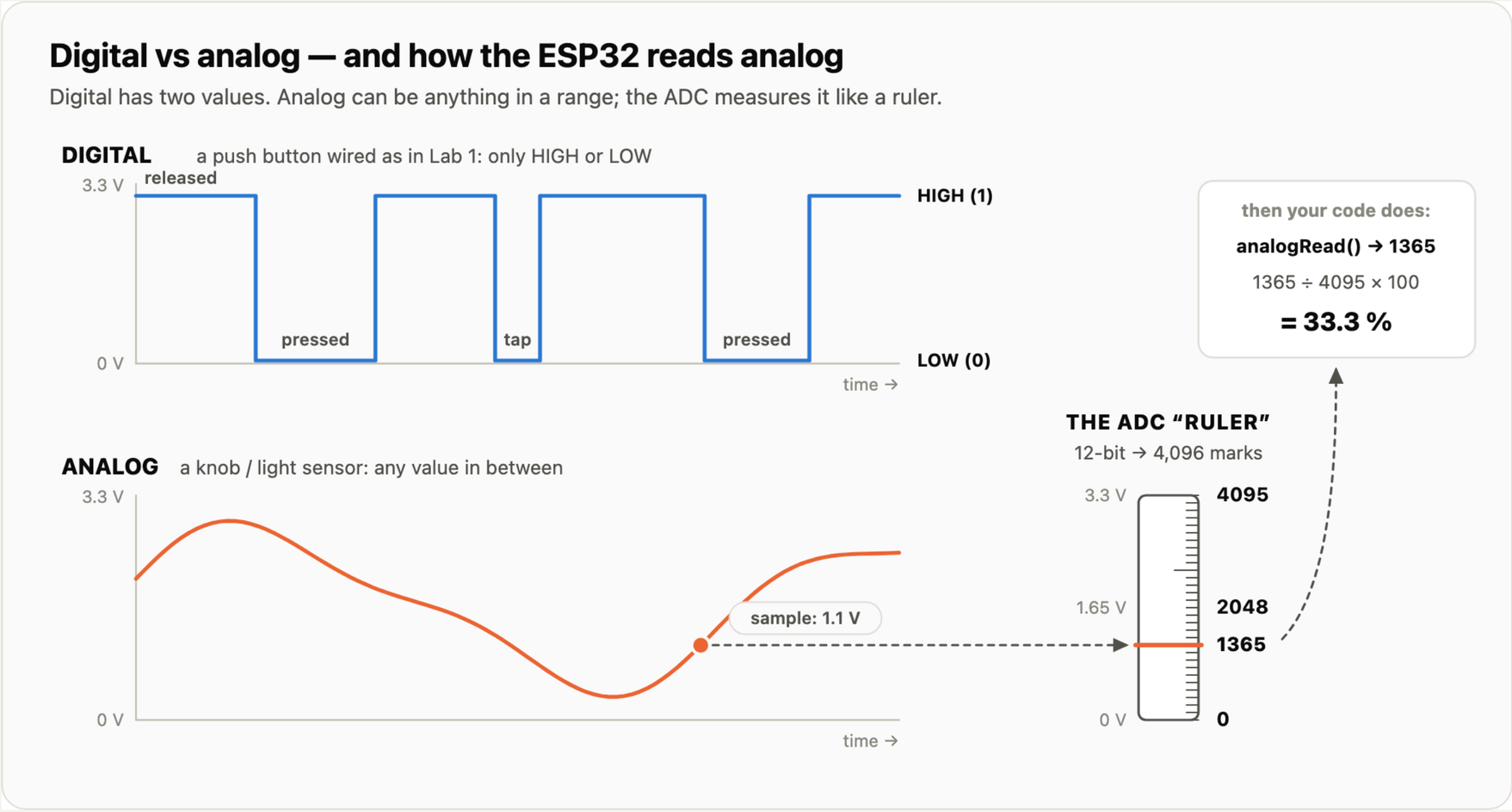
TWO KINDS OF SIGNAL

Digital and analog, side by side

	Digital	Analog
Values	Two: on/off, HIGH/LOW, 1/0	Any value in a range
Like a…	Light switch	Volume knob
Examples	Door contact, smoke alarm, button	Battery voltage, fuel level, dial
Read with	digitalRead()	analogRead()
It answers	“Is it happening?”	“How much?”

SEEING THE DIFFERENCE

From a voltage to a number



THE ESP32’S 12-BIT ADC

analogRead() gives 0 to 4095

0

at 0 V (nothing)

≈2048

at about 1.65 V (half-way)

4095

at about 3.3 V (full)

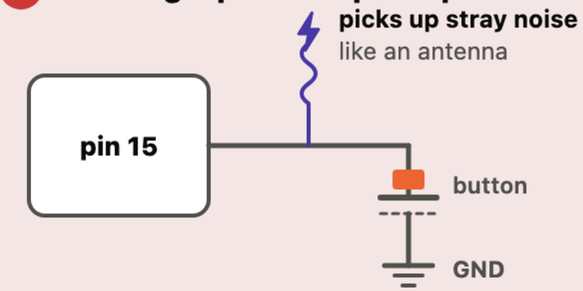
Read a raw number, then **convert** it: (raw ÷ 4095) × 100 = percent. With Wi-Fi, use **ADC1 pins (GPIO 32–39)**.

Don’t leave inputs floating

Why buttons need a pull-up resistor

An unconnected input “floats” and reads random noise. A pull-up holds it steady until pressed.

Floating input — no pull-up

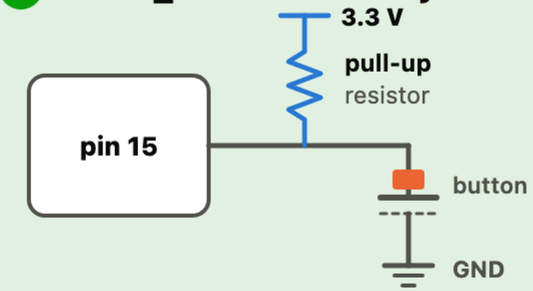


What digitalRead() returns, reading after reading:

H L H H L H L L H L H H L H

Random — the code sees presses that never happened.

INPUT_PULLUP — steady and reliable



What digitalRead() returns, reading after reading:

H H H H H L L L L H H H H H

button held down

HIGH at rest, LOW only while pressed.

Code: pinMode(15, INPUT_PULLUP);

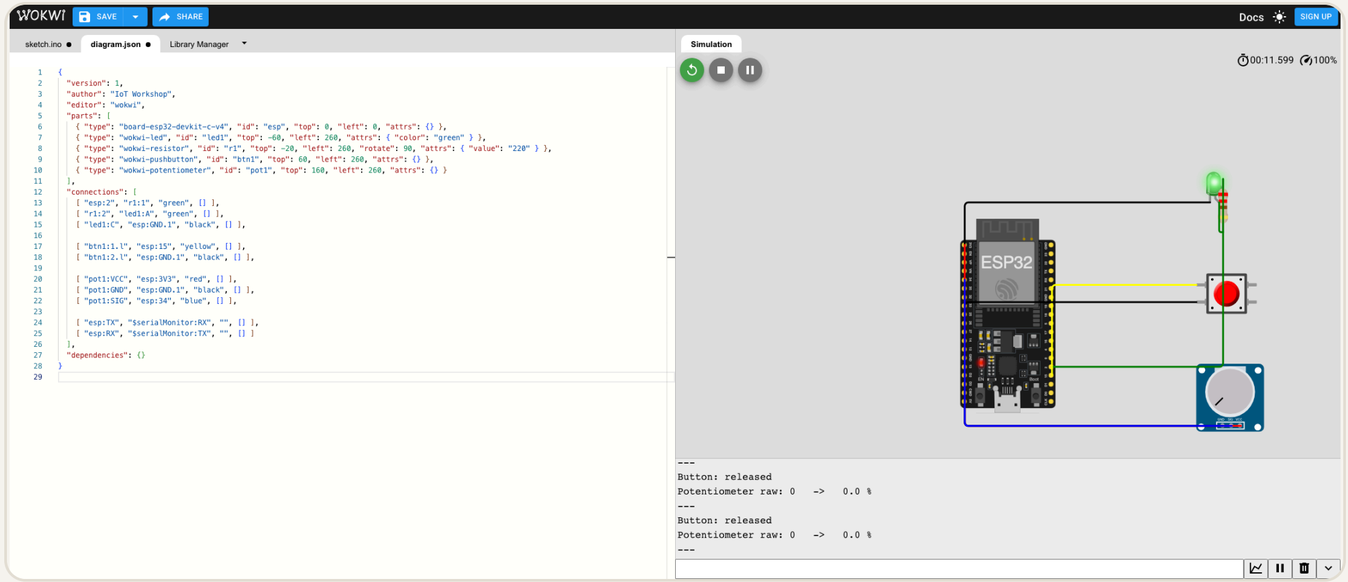
PAUSE AND CHECK

Your tower's shelter door alarm says "open" at random, but nobody is there. Why, and what's the fix?

Think about the picture we just saw.

LIVE DEMO · LAB 1 · 3 MINUTES

Watch: digital vs analog in Wokwi



1. I paste the code and circuit, press **Play**
2. The LED blinks: a **digital output**
3. I hold the button: **PRESSED**, a digital input
4. I turn the dial: **0 to 4095**, an analog input
5. Try it yourself later with the guide

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

IN TELCO OPERATIONS

What sensors change at a tower site

	Today	With IoT	The benefit
Hot shelter	Found at the next visit, or when equipment fails	The NOC gets an alert; extra cooling switches on	Faster fixes, fewer failures
Low battery	Found when the site goes down in a power cut	Battery voltage is watched; alert before it runs low	Fewer outages
Open door	Nobody knows until the next visit	The NOC gets a door alert at once	Less theft, safer sites

All three are **remote site monitoring**: fewer trips just to look.

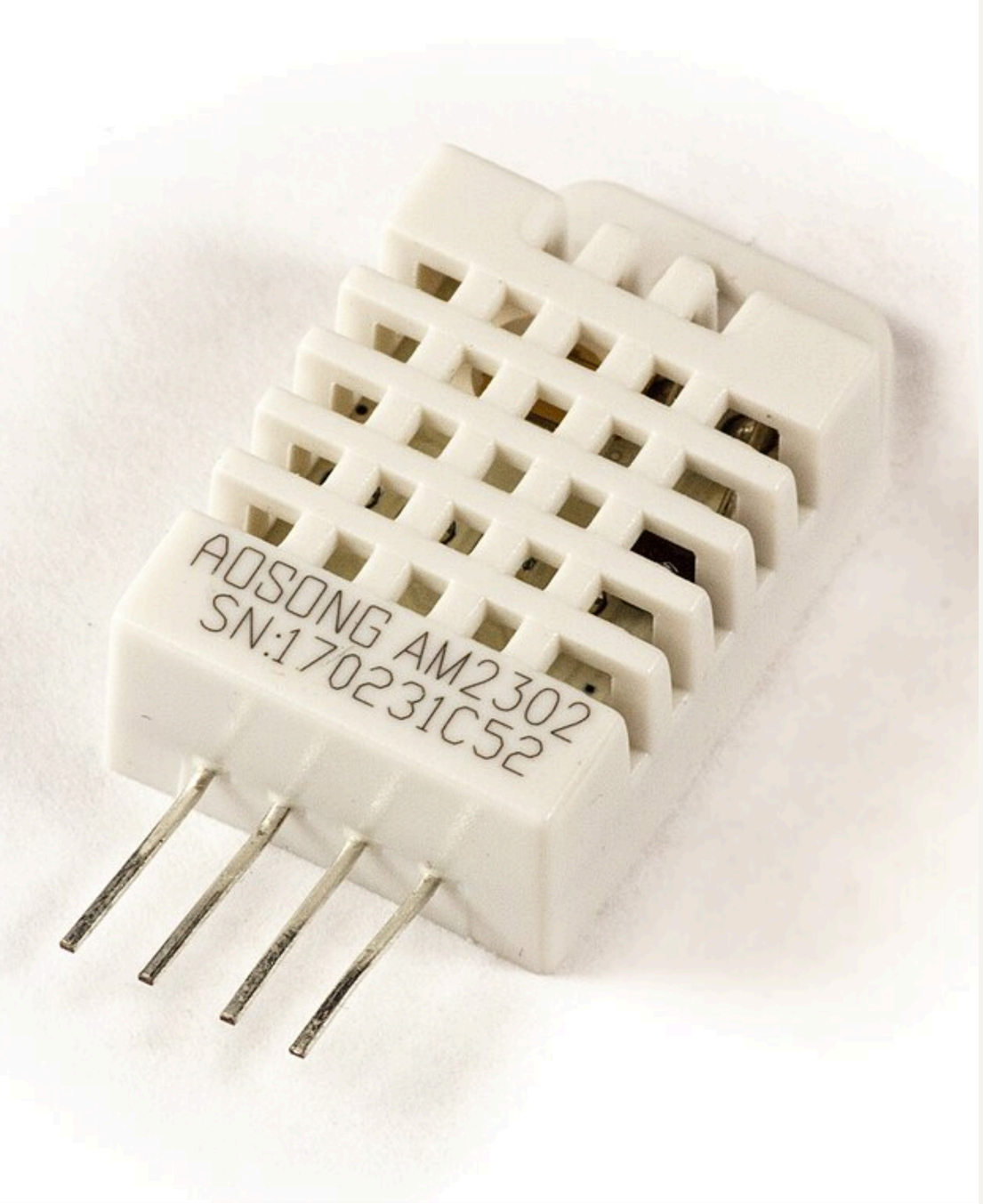


Photo: Ubahnverleih, CC0, via Wikimedia Commons

THE SENSOR FOR LAB 2

The DHT22

- Temperature **and** humidity in one small part
- A fresh reading only every **2 seconds**
- Accurate to about **±0.5 °C**
- Today: your **shelter** sensor



Photo: pslim, CC BY 2.0, via Wikimedia Commons

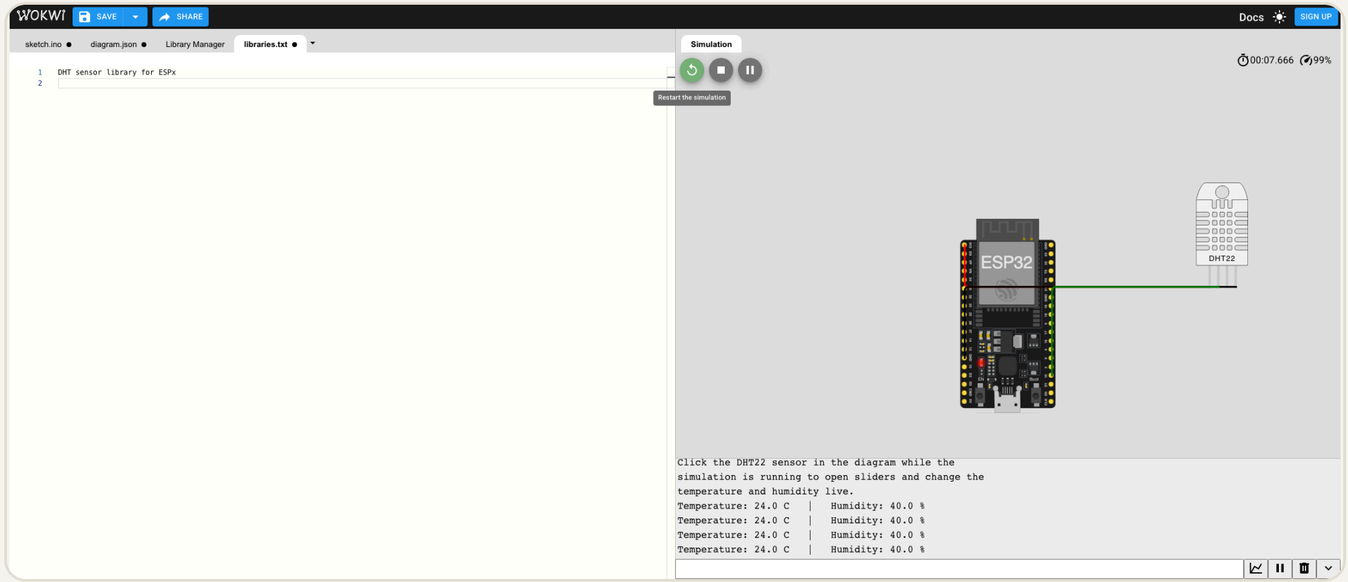
WHERE THE SENSOR GOES

Inside the equipment shelter

- A **shelter** holds the radios and batteries
- Nobody is there, so the **sensor** tells the NOC
- Too hot? Extra cooling **on at 28 °C**, off at 26 °C
- Which **five readings** from every site? **You decide**

LAB 2 · HANDS-ON · 20 MINUTES

Build the shelter sensor



1. Paste **sketch.ino** and **diagram.json**
2. Add **libraries.txt** with the DHT library name
3. Press **Play**: you see 24.0 °C and 40 %
4. Click the DHT22 and drag its sliders
5. Decision owner: which **five readings**?

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- **Sensors** turn heat, fuel level or an open door into signals
- **Actuators** act; a **relay** lets a small pin switch big equipment
- **Digital**: “Is it happening?” **Analog**: “How much?”
- The **ADC** gives 0–4095; the code **converts** it to real units
- Weigh the **sensor’s cost** against a **site visit** or an **outage**

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Name one digital sensor and one analog sensor at a tower site.

Q2 A fuel sensor’s analogRead() returns 1024. About how full is the tank?

Q3 Why can’t you connect the shelter’s cooling fan straight to an ESP32 pin?

Q4 The shelter’s DHT22 is read ten times a second. Why is that a problem?

UP NEXT · MODULE 4

Getting Online

How the data travels: Wi-Fi, cellular IoT and MQTT — and your shelter readings sent across the internet in Lab 3.

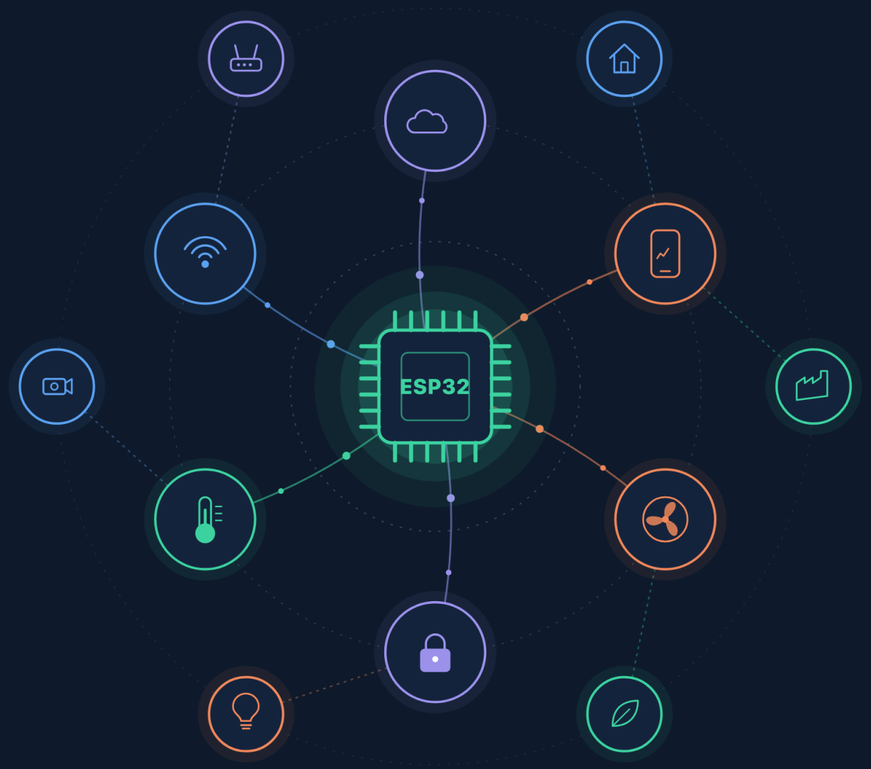
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 4 · 45 MINUTES

Getting Online: Connectivity and Communication Protocols

How a tower site sends its data to the NOC: the road, the language and what it is worth. Then Lab 3.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

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Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Separate** the road (connection) from the language (protocol)

02 **Choose** a connection for a job, by reach and monthly cost

03 **Explain** what connectivity is worth to a telco

04 **Explain** MQTT as a group chat, with topics

05 **Put** your Smart Site Monitor online in Lab 3

THE BIG PICTURE

Two separate decisions

Protocol

The **language** of the messages: MQTT or HTTP

Network

Addresses and routing: the internet

Connection

The **road**: mobile network, Wi-Fi, cable, Bluetooth

Lab 3: **Wi-Fi** is the road, **MQTT** the language. A tower site: the **mobile network** is the road.

Part A

Choosing the road

Which connection for which job, what it is used for, and what it is worth.

THE ROADS, IN PLAIN WORDS

Which connection for which job

Connection	Good for	A telco example	Running cost
Mobile-network IoT (NB-IoT, LTE-M)	Small readings from far away, on a small battery	Fuel, battery and door readings at a tower site	A monthly plan per site
Satellite	Sites with no mobile coverage	A very remote tower site	A higher monthly plan
Cable	Reliable, where cables exist	Data-centre meters and cooling	Free once installed
Wi-Fi	Lots of data, indoors	Cameras in an exchange or office	Free on our own network
Bluetooth	Tiny batteries, near a phone	A shelter lock opened by phone	Free; uses a phone

Ask: **how far**, **how much power**, **how much data**, and **what does it cost each month?**



Photo: pslim, CC BY 2.0, via Wikimedia Commons

REMOTE SITE MONITORING

A tower site, watched from the NOC

- **Sense:** fuel, battery, door, shelter heat
- **Road:** the mobile network, with a SIM
- **Language:** a light one, such as MQTT
- **Check:** fuel drops while the generator is off?
- **Cost:** one SIM plan, or one site visit?

TELCO USE CASES

Three jobs for mobile-network IoT



Remote sites

Fuel, battery, door and shelter heat, seen from the NOC



Smart towers

Thousands of towers, their power and alarms, on one NOC screen



Asset tracking

Trackers on generators, batteries, spares and vans

In our data centres and exchanges, **cable** and **Wi-Fi** do this job indoors, with no monthly plan.

THE BUSINESS LENS

What connectivity is worth

A SIM, not a trip

Compare a monthly SIM plan with one site visit, hours away

Coverage

The mobile network already reaches remote sites. No cable to lay.

Scale

One network and one NOC screen for thousands of sites

The cost is real, but **small and predictable**: per device, per month.

THE BUSINESS LENS

Telcos sell IoT connectivity too

1 billion

NB-IoT and LTE-M connections worldwide, end of 2025

3

Maxis IoT plans in Malaysia: NB-IoT, 4G M2M, 5G IoT

Each connected device can mean **a SIM and a monthly plan**: a cost for the buyer, revenue for the telco.

Source: GSMA (connections, end of 2025); Maxis Business website (plans).

PAUSE AND CHECK

Pick a road: (a) a fuel sensor at a remote tower site; (b) a tracker on a generator moved between sites; (c) a small battery sensor read by a nearby phone.

Ask: How far? Mains or battery? How much data? What is already there? What does it cost per month?

Part B

The language: MQTT

How one reading from a site reaches everyone who needs it.

MQTT · THE LANGUAGE OF LAB 3

MQTT works like a group chat



The site posts

The site sends one message: “shelter at 29 °C”



The broker

Like the chat group: it passes every message on



Everyone gets it

The NOC, the alarm system and the on-call phone

Lab 3: **Wi-Fi** is the road, **MQTT** the language. At a tower site, only the road changes.

MQTT TOPICS

Topics and wildcards

Which messages does a subscription receive?

Topics are levels separated by "/". + matches exactly one level. # matches everything below.

Message published to topic…	subscribed to: iot-workshop/+/temperature	subscribed to: iot-workshop/#	subscribed to: iot-workshop/aisha/#
iot-workshop / aisha / temperature	✓	✓	✓
iot-workshop / aisha / humidity	✗	✓	✓
iot-workshop / ravi / temperature	✓	✓	✗
iot-workshop / ravi / humidity	✗	✓	✗
iot-workshop / ravi / room1 / temperature	✗	✓	✗

two levels where + stands → no

IN TELCO OPERATIONS

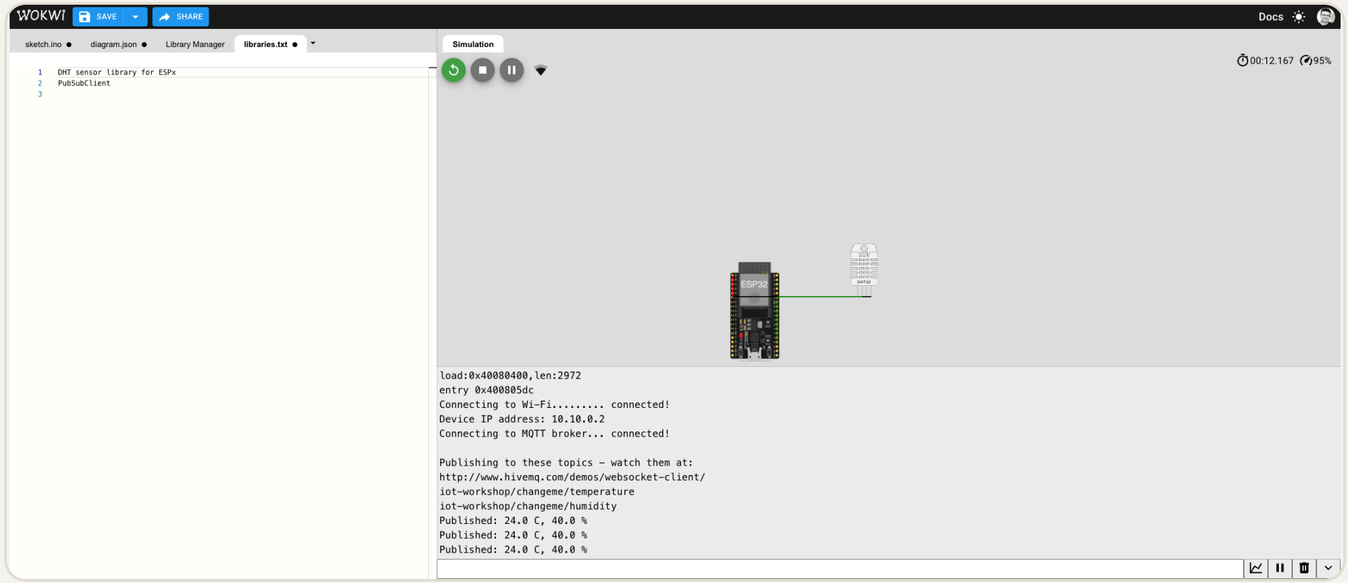
Getting sites online, in telco operations

Today	With IoT	The benefit
Site checks: drive out to look	Fuel, battery and door readings arrive by SIM	Fewer site visits
NOC alarms: passed on by phone	One message reaches the NOC, the alarm system and the on-call phone	Faster fault repair
Generators, spares: “where is it?”	Trackers show where it is, and if it moves	Less theft, less searching

Your team’s own numbers go on the canvas: costs in **box 6**, payback in **box 7** (Module 8).

LAB 3 · 23 MINUTES HANDS-ON

The tower site reports to the NOC



1. New builder: new ESP32 project in Wokwi
2. Paste sketch.ino — change “changeme” to your team name
3. Paste diagram.json and add libraries.txt
4. Press Play: Wi-Fi, then MQTT, then readings
5. Watch in HiveMQ, then fill in canvas box 3

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- Two decisions: **the road** (connection) and **the language** (protocol)
- Pick the road by the job: **mobile-network IoT** at remote sites
- Connectivity pays when **a SIM plan replaces a site visit**
- Telcos **sell** IoT connectivity: 1 billion NB-IoT and LTE-M links
- **MQTT** is a group chat: the site posts once, everyone subscribed gets it

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 A remote tower site has no Wi-Fi. How can its fuel sensor send data?

Q2 What is the difference between the road and the language?

Q3 A site publishes one MQTT message. Who receives it?

Q4 Does towers/+/temp match towers/north/site7/temp?

UP NEXT · MODULE 5

IoT Cloud Platforms and Data Visualisation

We give the site’s data a home: a live cloud chart in Lab 4, using HTTP this time.

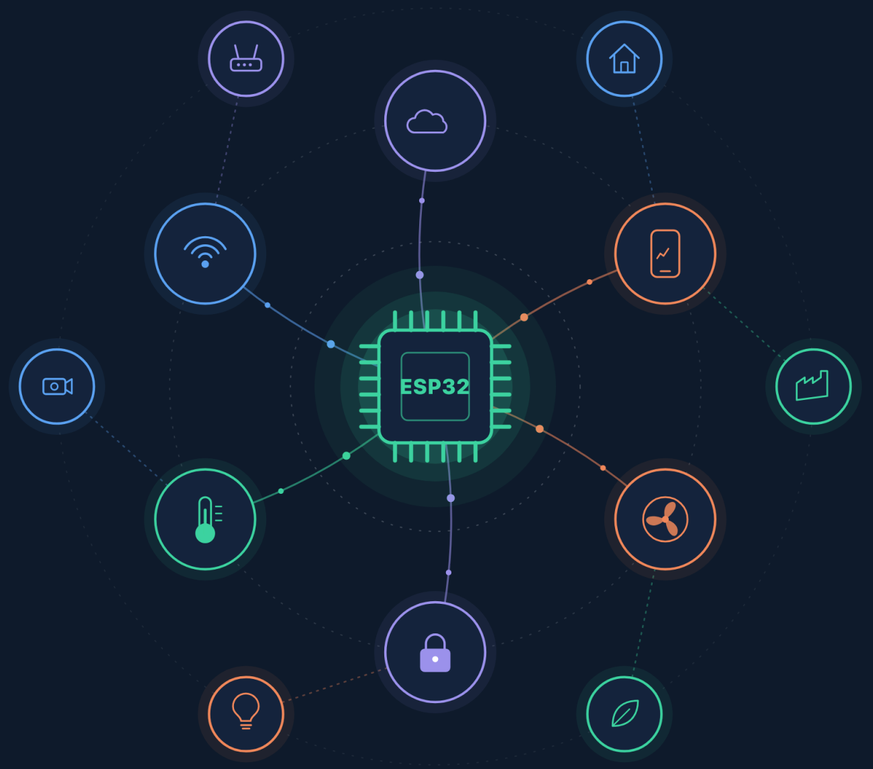
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 5 · 45 MINUTES

IoT Cloud Platforms and Data Visualisation

Where a tower site’s readings go, and how to build dashboards for the NOC and for managers.



Mehdi Lotfinejad

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BY THE END OF THIS MODULE

You will be able to...

01 **Explain** what an IoT cloud platform does with site data

02 **Design** two views: one for the NOC, one for managers

03 **Choose** the right chart for each question

04 **Describe** an energy view: diesel, energy mix and PUE

05 **Build** a live trend view of a site’s shelter in Lab 4

Part A

From the tower site to the cloud

In Lab 3, your readings vanished when you closed the tab. Where should they live?

FROM THE SITE TO THE DASHBOARD

What an IoT platform does

Ingest

Readings arrive over the network.

Store

Every reading is kept with its time.

Analyse

Averages, trends and alerts.

Visualise

Charts and dashboards for people.

Rules and actions

“If this, then that”: in Module 6.

The cloud means renting computers and storage over the internet. It runs in data centres.

THE PLATFORM FOR LAB 4

ThingSpeak: channels, fields and keys

Channels

One per project or device, like “Tower Site 12”.

Fields

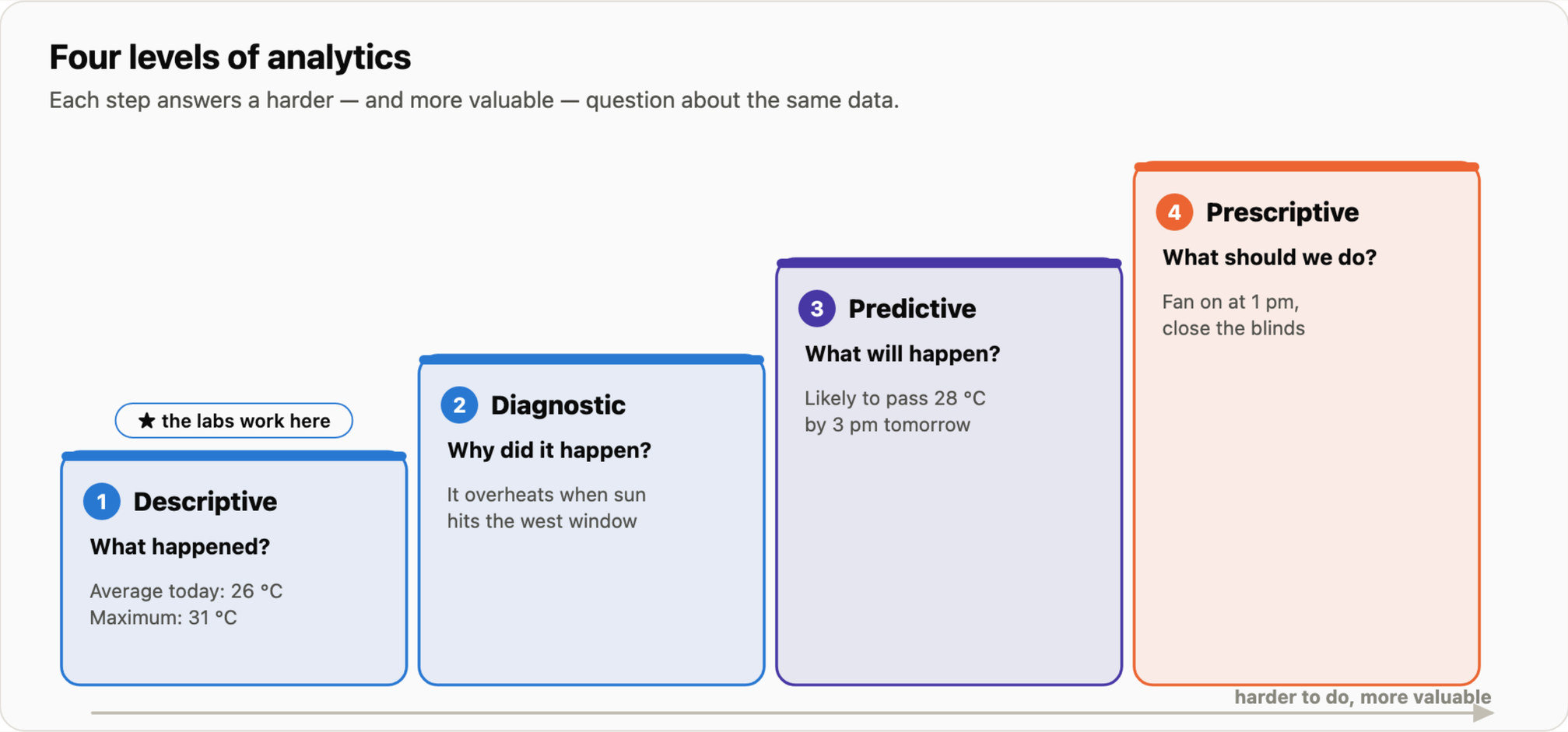
Up to 8 values per channel: Field 1 = Temperature.

API keys

The **Write** key adds data. The **Read** key reads a private channel.

Free plan: about 3 million messages a year, 4 channels, **15 s** between updates, non-commercial

Four kinds of analytics



Part B

Dashboards for two audiences

One set of data, two very different screens.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

SMART TOWER MONITORING

One NOC screen, many sites

- **One screen** for hundreds of tower sites
- **A status light per site:** OK, warning or alarm
- **Alarms now:** shelter hot, door open, generator on
- **Worst first**, with a word, not only a colour
- **Action:** call the site, or send an engineer

SAME DATA, TWO SCREENS

Two dashboards for two audiences

	Operations (NOC)	Management
Question	Is any site in trouble **now**?	Are we getting **better**?
Time range	The last hour	Last month, last year
Headline	Alarms: shelter hot, door open	Energy cost, diesel, uptime
Best visual	Status lights and gauges	Big numbers and line charts
Action	Send a field engineer	Approve a budget, set a target

Rule one of every dashboard: **know your audience**.

CHOOSING THE RIGHT CHART

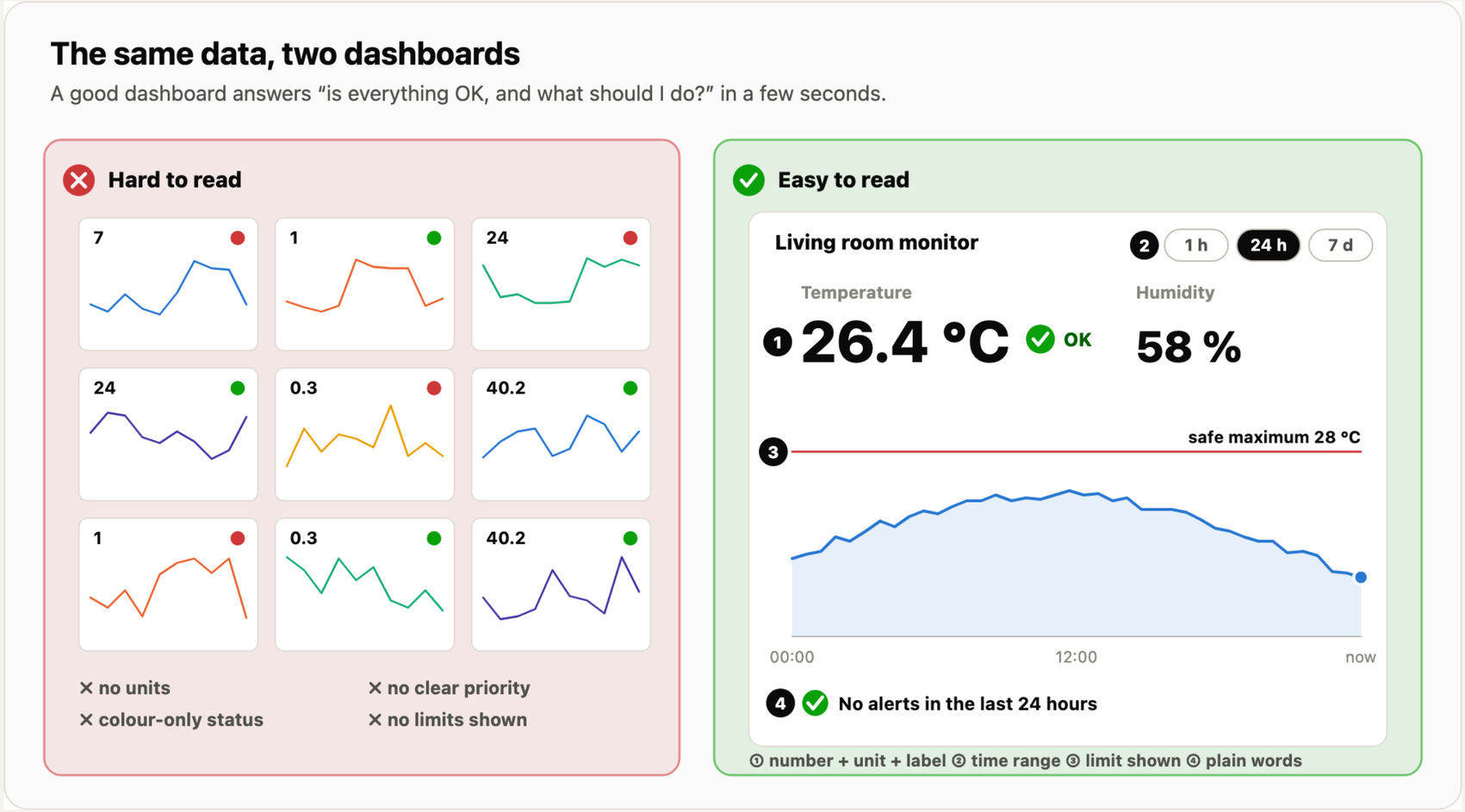
Different questions need different visuals

To show…	Use a…	Example
Change over time	Line chart	Shelter temperature, last 24 hours
Value right now	Gauge or big number	“Fuel tank is 72% full”
A comparison	Bar chart	Diesel used by each site this month
On/off, good/bad	Status indicator	“Generator: RUNNING”
Where things are	Map	Tower sites in the region
Exact values	Table	The last 10 readings, with times

For IoT sensor data, the **line chart** is the workhorse.

DASHBOARD DESIGN

Good dashboard design in action



PAUSE AND CHECK

A hospital wants one screen showing whether each of 20 medicine fridges is in its safe range. Which visual would you choose, and what would you avoid?

Think of the NOC view. Discuss in your team for 30 seconds.

Part C

Energy: the view managers need

Diesel, power and cooling are some of the biggest bills in the network.

ENERGY MANAGEMENT

The energy view of the tower network

Diesel

Litres used per site, per month

Energy mix

Grid, battery, generator and solar, per site

Run-hours

Generator hours, compared with the fuel level

Hybrid sites

EDOTCO: hybrid renewable power for off-grid Sabah sites

Energy is **15–20%** of a mobile operator’s operating costs. Sources: GSMA Intelligence, 2025 · EDOTCO, 2019

ENERGY IN OUR OWN DATA CENTRES

PUE: how efficient is the data centre?

1.56

average PUE in 2024, flat for five years

>30%

of power on cooling in less efficient sites

PUE = all the site's electricity ÷ the IT equipment's electricity. **1.56:** for every 1 kWh of IT, another 0.56 kWh goes on cooling and losses.

Source: Uptime Institute Global Data Center Survey 2024 (PUE) · IEA, *Energy and AI*, 2025 (cooling)

THE BUSINESS LENS

Which three numbers would you show the board about the tower network?

Think about money, risk and customers. Start now, and decide in the lab.

IN TELCO OPERATIONS

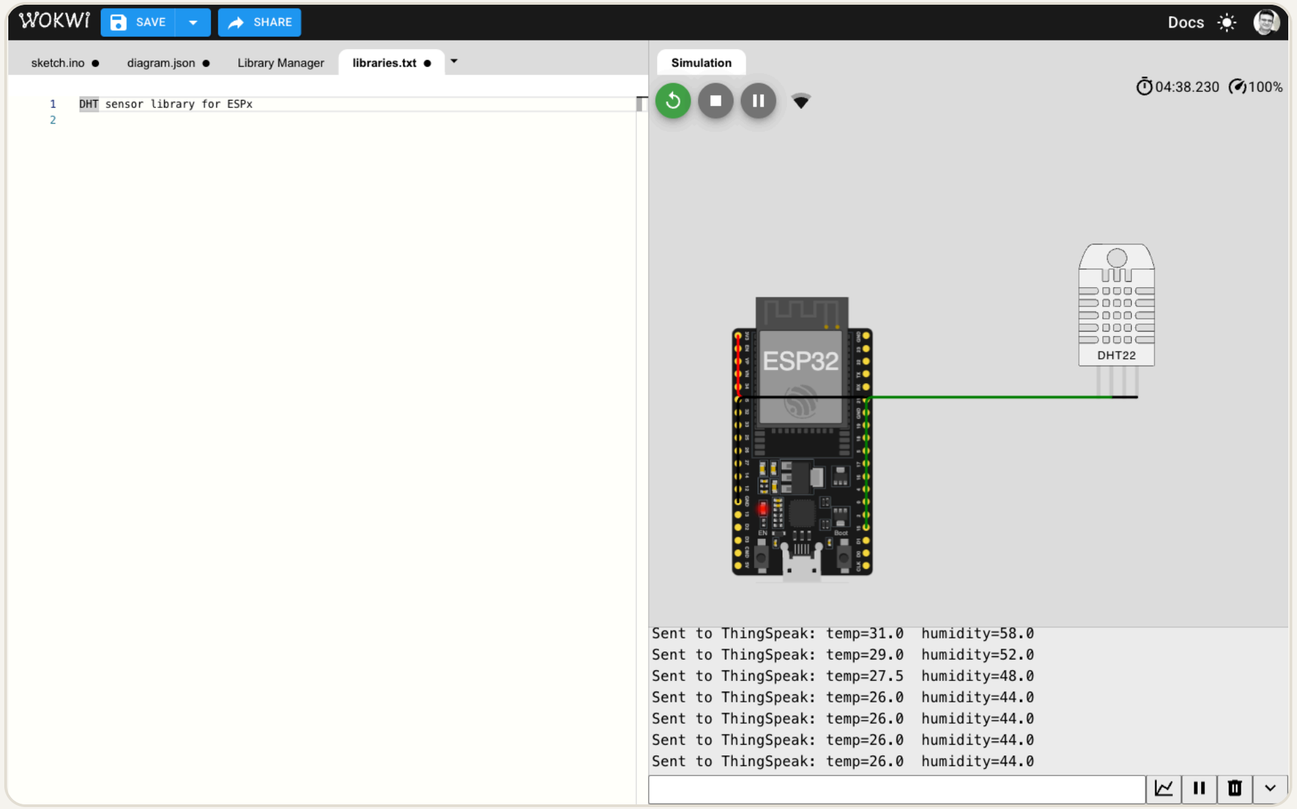
Cloud dashboards in telco operations

	Today	With IoT	The benefit
NOC alarms	Many alarms: which matter?	A status light per site	Faster fault repair
Field visits	An engineer drives out to look	Check the trend; go if needed	Fewer site visits
Site energy	Diesel seen on visits and bills	Diesel and energy mix per site	Less diesel, less fuel theft

Read each row: **today → with IoT → the benefit**

LAB 4 · 22 MINUTES HANDS-ON

Lab 4: the trend view of one tower site



1. Create a ThingSpeak channel with two fields
2. Copy your **Write API key**
3. Paste the key into Wokwi and press Play
4. Watch the charts grow every 20 seconds
5. **Manager task:** 3 numbers for a board report

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- An **IoT platform** receives, stores, analyses and shows site data
- **ThingSpeak:** channels, fields, and a Write key you keep secret
- **The NOC view:** a status light per site, worst problems first
- **The energy view:** diesel, energy mix and run-hours per site
- **Line charts** for change over time; status lights for “right now”
- **PUE 1.56:** in data centres, 0.56 kWh extra for every 1 kWh of IT

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 The NOC watches 500 tower sites. What should its screen show first?

Q2 Name two things an energy view shows for each tower site.

Q3 Which chart shows how a shelter’s temperature changed last week?

Q4 In ThingSpeak, what does the Write key do? And the Read key?

UP NEXT · MODULE 6

Data Processing and Automation

The Smart Site Monitor starts to act: extra cooling in the shelter switches on at 28 °C and off at 26 °C, with no cloud needed.

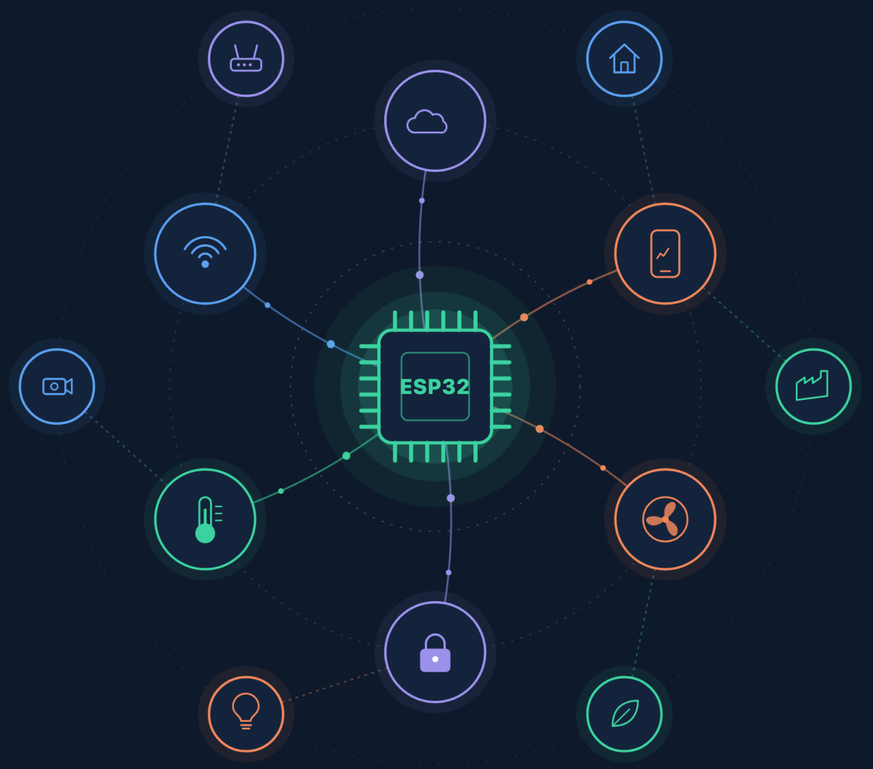
Mehdi Lotfinejad

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MODULE 6 · 45 MINUTES

Data Processing and Automation

How a system turns raw readings into decisions, and then acts on them by itself.



Mehdi Lotfinejad

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BY THE END OF THIS MODULE

You will be able to...

01 **Describe** how raw readings become a decision and an action

02 **Write** simple IF-THEN rules for a tower site

03 **Explain** hysteresis, and why edge rules work offline

04 **Decide** what runs automatically and what needs a person

05 **Build** a site monitor that switches on cooling by itself

THE BIG PICTURE

From reading to action



Raw data

31.4, 31.6, 31.5 °C in the shelter



Process

Check, clean and average



Rule

Is it above 28 °C?



Act or wait

Yes: cooling on + alert. No: keep watching.

Module 1’s loop: **sense** → **send** → **decide** → **act**. Today is **decide** and **act**.

AUTOMATION RULES: IF THIS, THEN THAT

Four automations for a tower site



Hot shelter

IF above 28 °C, **THEN** extra cooling + an alert.



Battery low

IF the battery is low, **THEN** start the generator.



Fuel theft?

IF fuel drops, generator off, **THEN** alert the NOC.



Service early

IF battery health falls, **THEN** book a service.

Energy is **15–20%** of a mobile operator’s operating costs (GSMA Intelligence, 2025).

PREDICTIVE MAINTENANCE

Service it before it fails

Battery health

The trend falls week by week: replace it in time.

Run-hours

The generator reaches its service hours: book a visit.

Cooling units

It works harder for the same temperature: check it.

Not on a fixed calendar, not after it breaks: **when the data says so**.

THE SAME IDEA IN OUR DATA CENTRES

Power failures cause the worst outages

No. 1

cause of serious outages: power

1 in 5

operators: last major outage cost over US$1 million

UPS systems, transfer switches and generators fail most often. Watch their health and fix them early.

Source: Uptime Institute, Annual Outage Analysis 2026

A FAMOUS EXAMPLE

DeepMind: less cooling energy from old data

Up to 40%

less energy used for cooling

1,000s

of sensors the sites already had

Machine learning on data the sensors **already collected**: temperatures, power, pump speeds.

Source: Google DeepMind blog, July 2016

Automated network fixes: the same pattern



Network data

Alarms, traffic and faults from the network itself.



Decide

Software spots a problem or a pattern.



Act

It changes network settings, or calls a person.

This is **network automation**, not IoT: no new devices. But the pattern is the same: **decide → act**.

WHY CLEANING MATTERS

One bad reading can cause a false alarm



Validate

A loose wire sends **85 °C**. Rule: ignore a jump of more than 10 °C.



Smooth

Near the limit: 27.9, 28.1, 27.8, 28.2. Average a few first.

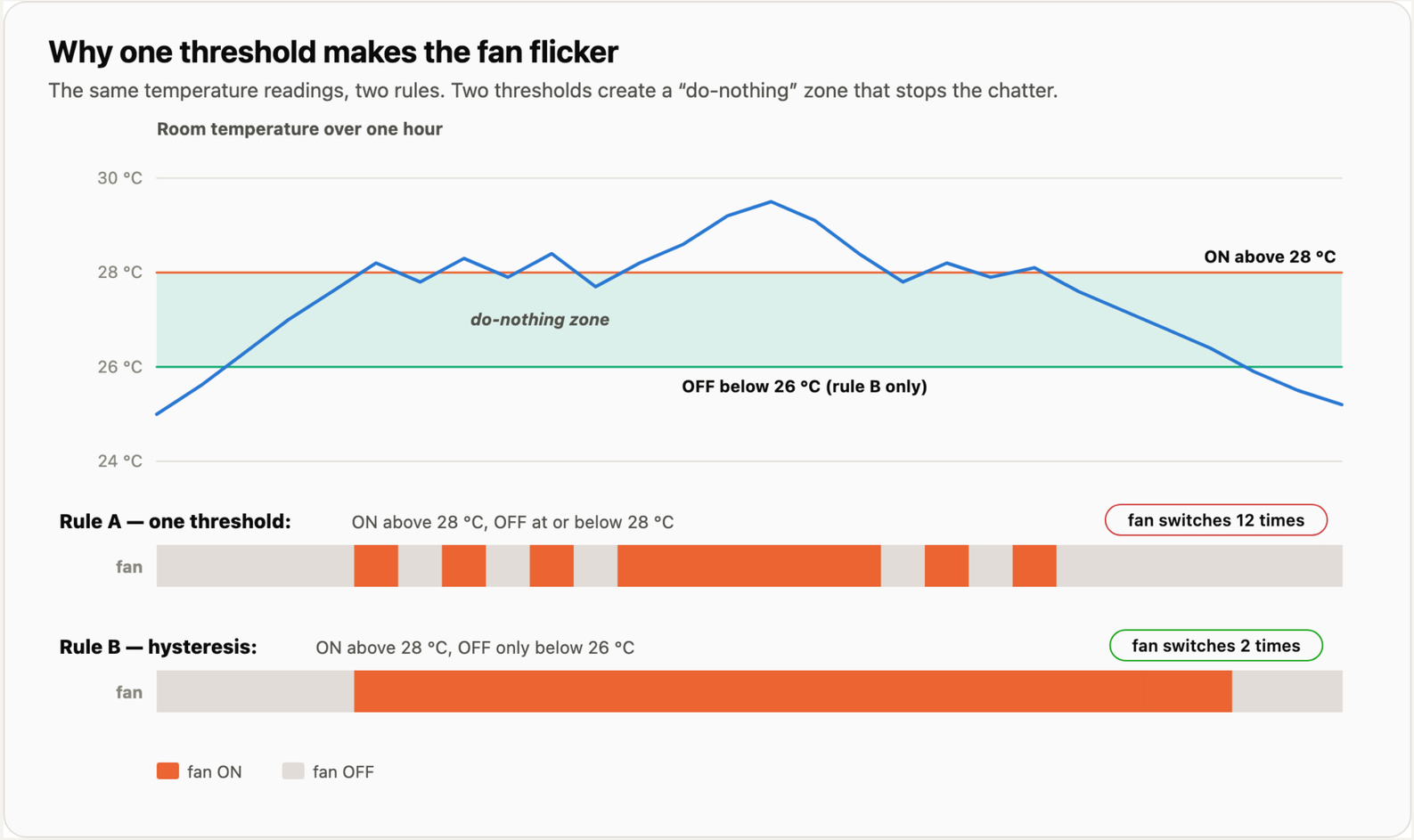
PAUSE AND CHECK

Your shelter sensor reports 24.1, 24.3, 24.2, 97.0, 24.2 °C. Which step should deal with the 97.0?

Validation or smoothing? Think on your own for 20 seconds.

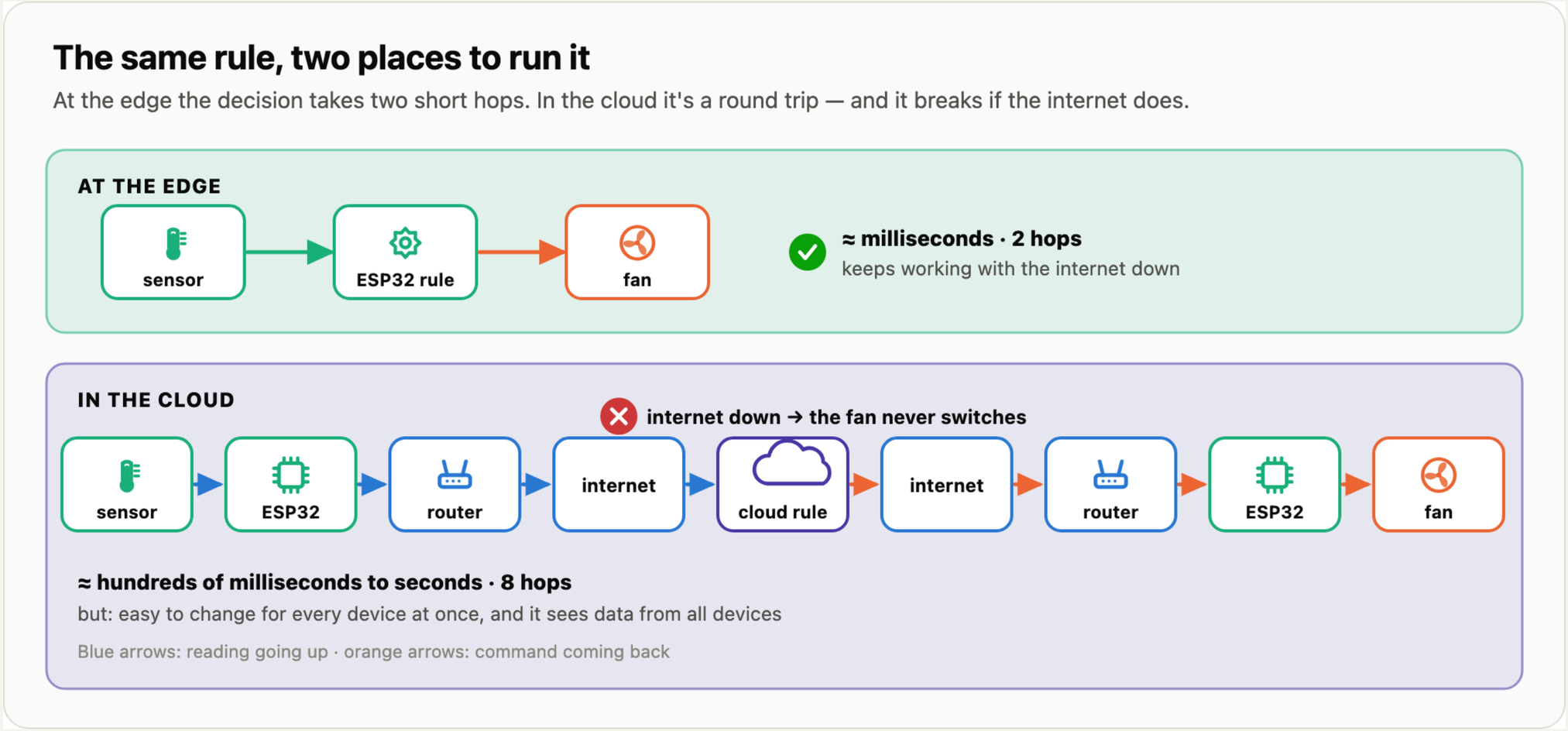
A CLASSIC RULE PROBLEM

Flickering, and the fix: hysteresis



EDGE VS CLOUD

The same rule, two places to run it



THE BUSINESS LENS

What a false alarm costs

Wasted trips

A long drive to a tower, for nothing.

Lost sleep

The on-call engineer is woken at 3 am.

Lost trust

Too many alarms, and people ignore the real one.

Wrong action

Good equipment is switched off.

Fewer false alarms: **clean** the data, use **hysteresis**, add a **time delay**.

DECIDE → ACT

Automatic, or a person confirms?

Situation	Action	Who acts?
Shelter above 28 °C	Switch on extra cooling	**Automatic**
Battery low, no mains	Start the generator	**Automatic**
Battery health falling	Book a service visit	**Person confirms**
Fuel drops, generator off	Send a team to the site	**Person confirms**
Data hall: three racks hot	Move or shut down servers	**Person confirms**

Automatic when it is **fast, safe and easy to undo**. A person confirms when it is **costly, risky or hard to undo**.

IN TELCO OPERATIONS

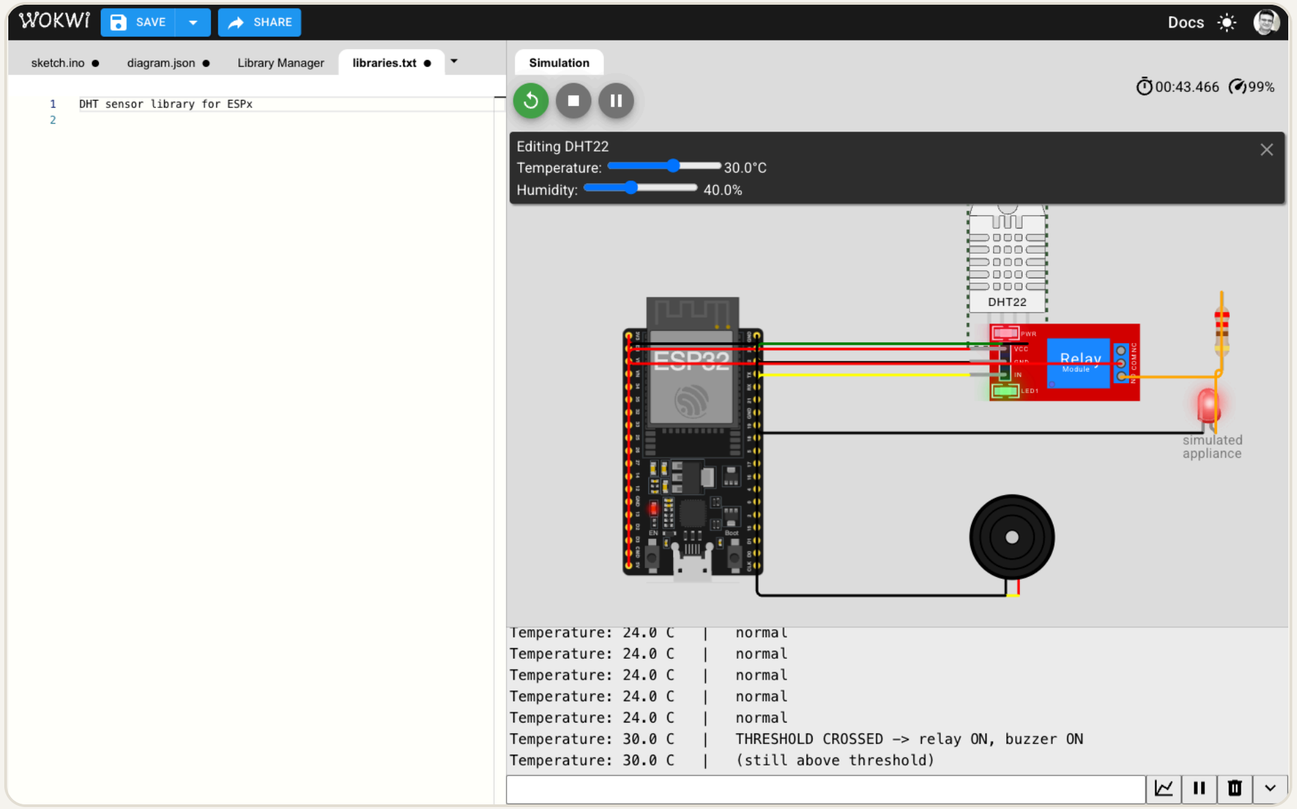
What automation changes at our sites

Today	With IoT	The benefit
Hot shelter: an engineer drives out	Cooling on by itself; NOC told	Faster fix, fewer site visits
Diesel: the generator runs too long	Runs only when the battery needs it	Less diesel, less wear
Maintenance: by calendar, or after failure	Service when the data says so	Fewer outages, better use of engineers

Canvas **box 4**: which row is closest to your team’s problem?

LAB 5 · 22 MINUTES HANDS-ON

Hot shelter: automatic cooling and alarm



1. Paste sketch.ino, diagram.json and libraries.txt
2. Press **Play**: a reading every 2 s, all “normal”
3. Drag the DHT22 above 28 °C: the shelter is hot
4. Relay = extra cooling **ON**; the buzzer alerts
5. Fill in canvas **box 4**: the decision and the action

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- A rule turns data into action: **IF** condition **THEN** action
- **Clean** the data first: one bad reading can cause a false alarm
- **Hysteresis** (ON at 28 °C, OFF at 26 °C) stops flickering
- **Edge** rules work offline; the **cloud** sees every site
- Site rules pay: cooling, generator, fuel theft, early service
- Automate what is safe and easy to undo; a **person confirms** the rest

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Why does the shelter cooling switch ON at 28 °C but OFF only at 26 °C?

Q2 The link to a tower site is down. Where should the cooling rule run?

Q3 Fuel drops overnight while the generator was off. What should happen?

Q4 Which two readings tell you to service a site before it fails?

UP NEXT · MODULE 7

IoT Security and Best Practices

What can go wrong: how IoT devices get attacked, and how to protect tower sites, data centres and the network.

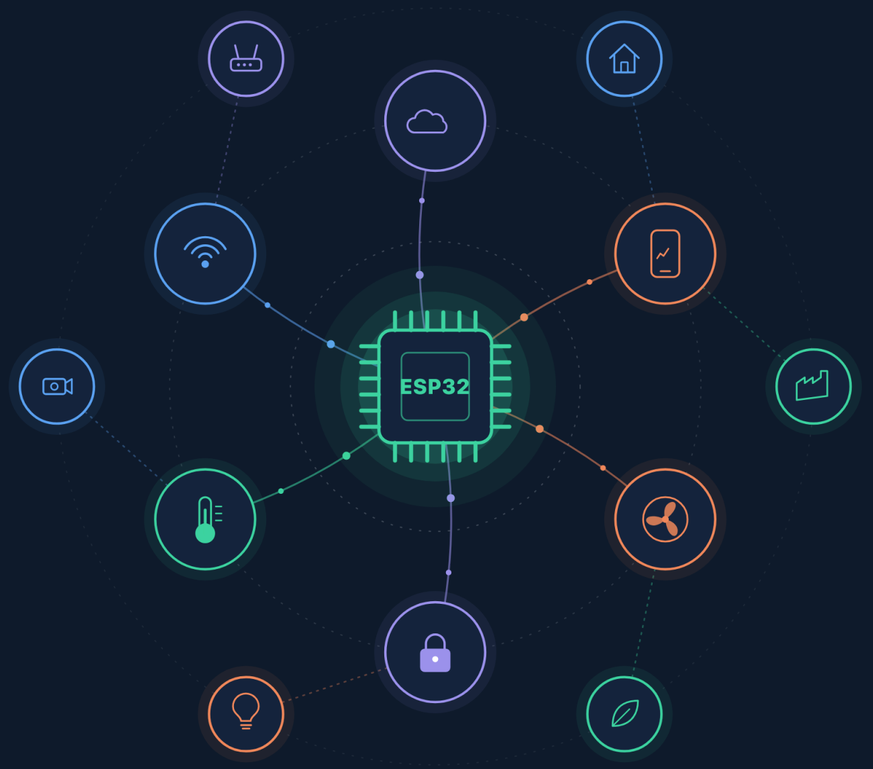
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 7 · 45 MINUTES

IoT Security and Best Practices

Why a cheap sensor at a remote site can be the way in, the simple defences, and what to ask a vendor.



Mehdi Lotfinejad

Introduction to the Internet of Things

THE ROAD AHEAD

Eight modules, one Smart Site Monitor

PART 1 — DEVICES AND DATA

01
What is IoT?
Lecture

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How it's built
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PART 2 — CLOUD, AUTOMATION AND SECURITY

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Cloud and dashboards
Lab 4

06
Automation
Lab 5

07
Security
You are here

08
Real world + pitch
Lab 7

BY THE END OF THIS MODULE

You will be able to...

01 **Explain** why a cheap sensor at a remote site can be the way in

02 **Describe** simple defences: identity, separate networks, encryption

03 **Decide** who owns vendor risk, and what to ask before buying

04 **Review** the Smart Site Monitor before it goes onto tower sites

05 **Fill in** box 8 of your business-case canvas: the risks

Part A

How attackers get in

The tower site first, then two true stories.

OUR MAIN CASE · THE TOWER SITE

Security at a remote tower site

Shelter door

Who opened it, and when? Often nobody knows until the next visit.

Remote access

Contractors log in to the tower RMS from their own office.

Fuel and batteries

Diesel and batteries are valuable, and sites are far away.

Now we add a new device to every site: the IoT monitor. Is it a guard, or **another door**?

FOUR REASONS ATTACKERS LIKE IOT

Why IoT devices are easy targets

So many

Billions of devices, and many are identical.

Rarely updated

A weakness found in year one is still there in year five.

Rushed design

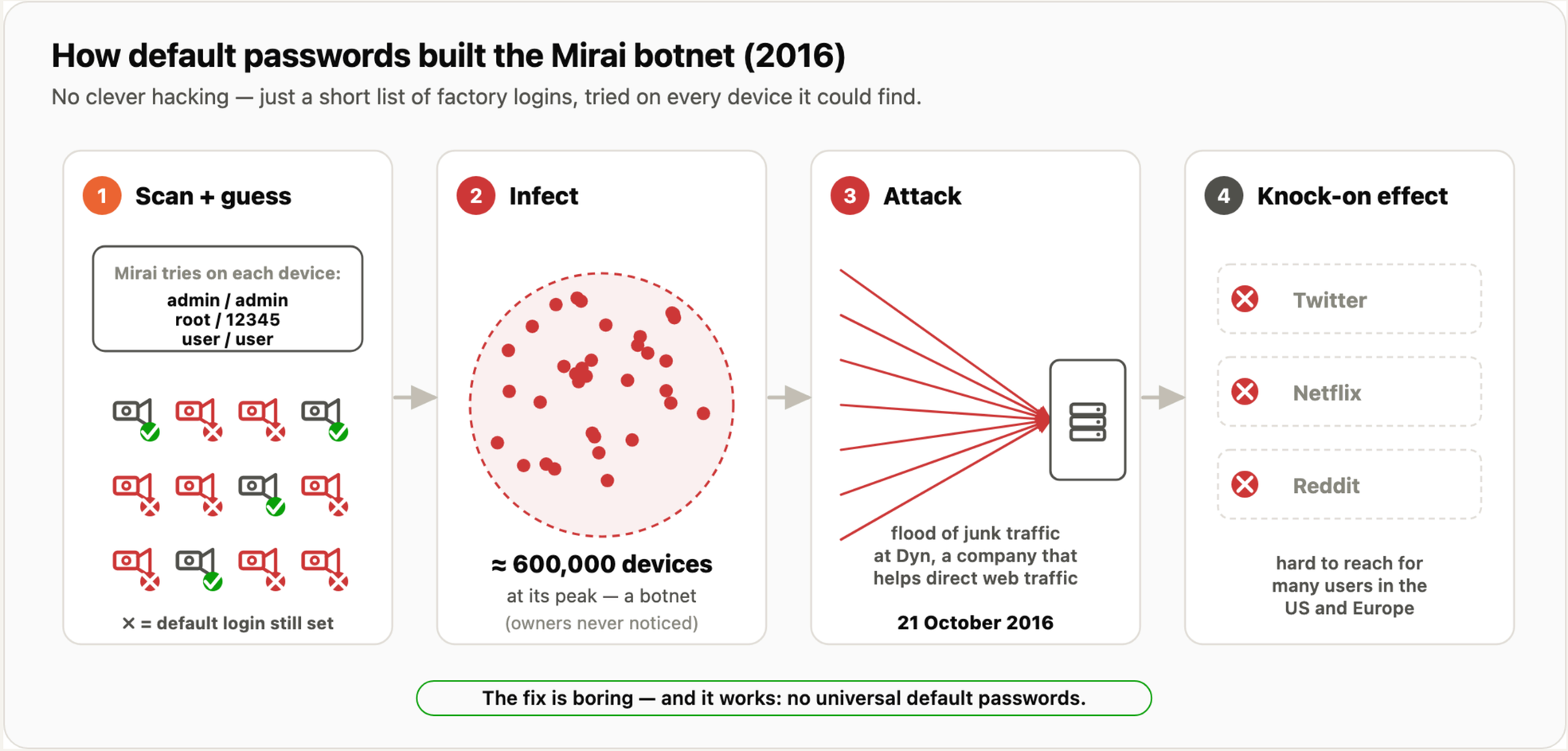
Weak passwords, old software, and no way to fix them.

Physical world

An attack can open a door, or hide a fuel theft.

TRUE STORY 1 · 2016

The Mirai botnet



TRUE STORY 2 · 2013

The Target breach



The contractor

A heating and cooling (HVAC) firm had a login to Target’s network.



The theft

Attackers stole that login from the contractor.



The way in

The breach began with that stolen login.

The weak point was a **supplier**, not Target itself. Source: Krebs on Security, 2014.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

THIRD-PARTY RISK

Who else has a key to your site?

- The tower RMS provider
- The generator and battery maintenance team
- The fuel delivery contractor
- The cooling vendor in our data centres
- The sensor maker’s cloud

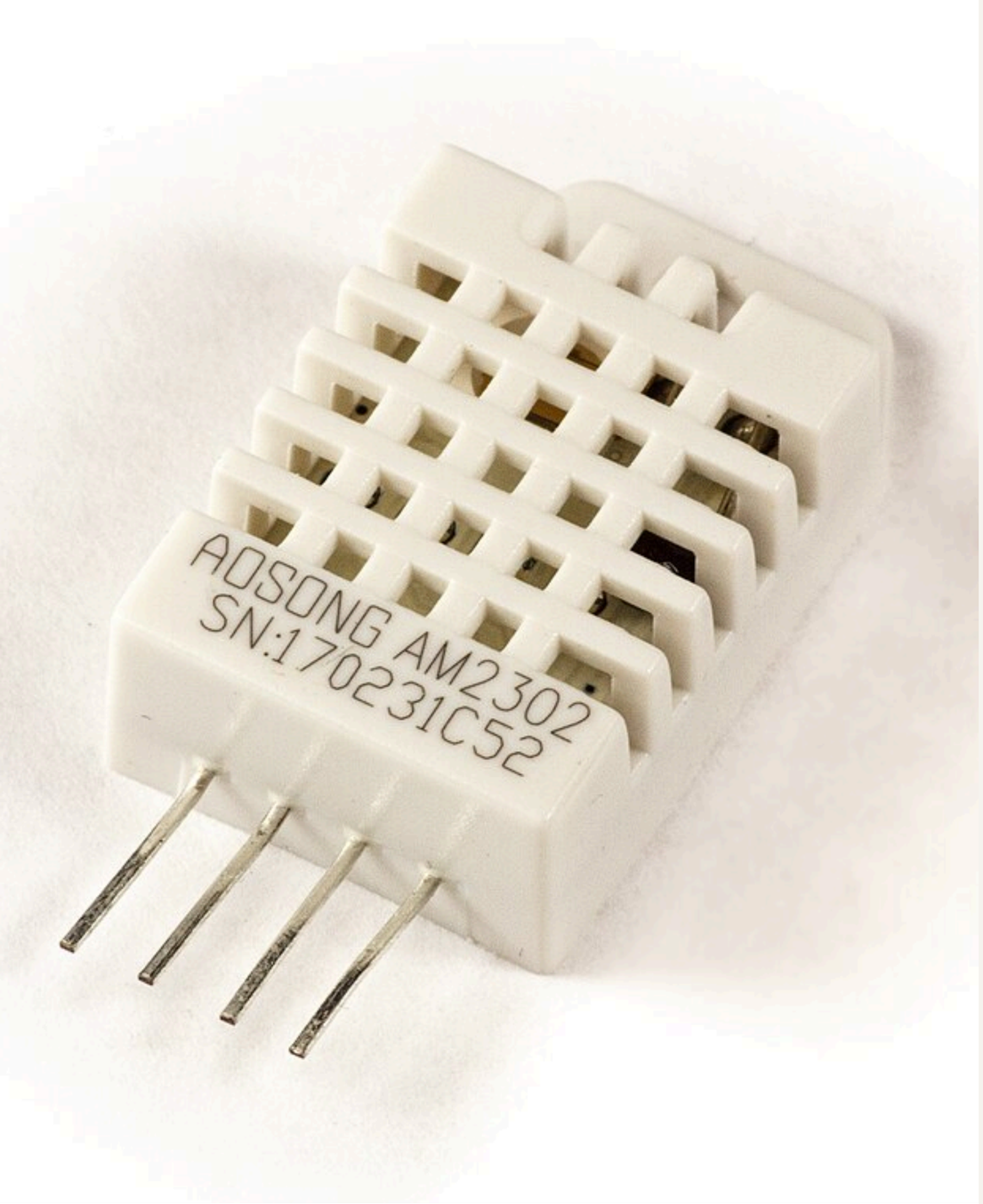


Photo: Ubahnverleih, CC0, via Wikimedia Commons

THE KEY IDEA TODAY

A cheap sensor can be the way in

- A sensor is a small computer on your network
- It often ships with a default password
- Nobody updates it, and nobody watches it
- From there, an attacker can reach other systems
- A fake reading can hide a hot shelter

PAUSE AND CHECK

A cheap sensor in the shelter at Tower Site 12 still has its factory password. Why is that a risk for the whole network?

Think of Mirai and Target. Talk to your neighbour for one minute.

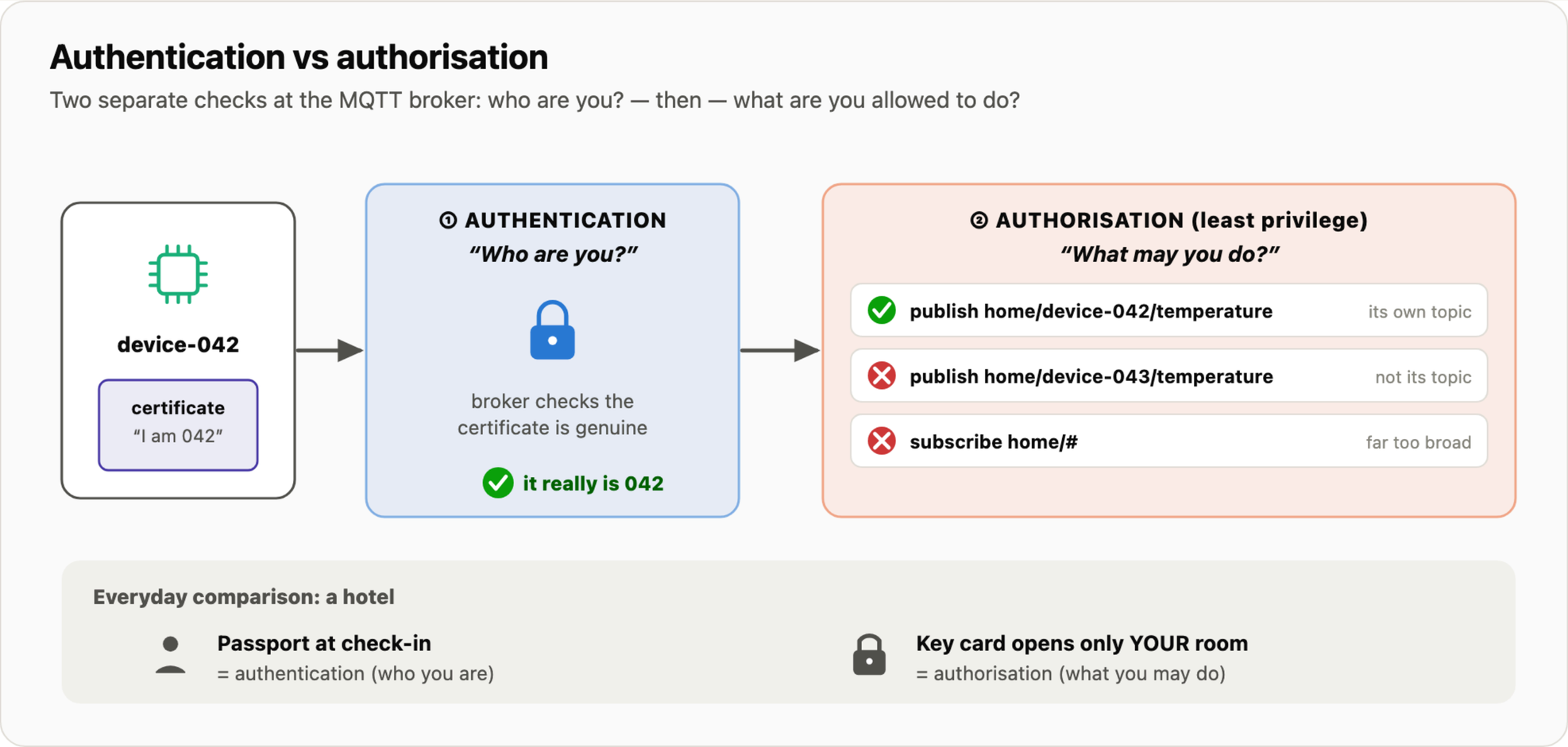
Part B

Simple defences

Strong identity, separate networks, encryption and updates.

DEFENCE 1 · IDENTITY

Authentication and authorisation



DEFENCE 2 · SEPARATE NETWORKS

Keep site systems off the office network

Office IT

Email, laptops and finance. Many users, many risks.

Site systems (OT)

Tower RMS, batteries, generators, doors; cooling and the BMS.

One guarded gate

A firewall lets only the traffic you need pass.

If a sensor is attacked, the attacker is stuck in one **zone**. The industrial standard IEC 62443 uses zones like this.

DEFENCES 3 AND 4 · PASSWORDS, ENCRYPTION, UPDATES

Securing the Smart Site Monitor

In the labs	At a real tower site
Public MQTT broker, no password	Our own broker; a unique login or certificate per device
Port 1883, not encrypted	Port 8883, encrypted with TLS
Any client can read iot-workshop/#	Each monitor uses only its own topics
Plain HTTP to ThingSpeak	HTTPS
Wi-Fi password written in the code	Secrets kept separately, never shared
No way to update the device	Signed updates, with a stated support period
On any Wi-Fi network	On the separate site (OT) network, via 4G

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ **Myth** *Our sensors are too small and boring to attack.*	✓ **Fact** Tools like Mirai scan the whole internet for any weak device.
✗ **Myth** *Security is the vendor’s problem.*	✓ **Fact** It’s shared. We must change passwords, update, control access.
✗ **Myth** *Encryption solves everything.*	✓ **Fact** It doesn’t stop weak passwords, fake updates or stolen keys.
✗ **Myth** *Once it’s secure, it stays secure.*	✓ **Fact** New weaknesses appear all the time. Devices need updates.

Part C

The business lens

Who owns the risk, what to ask a vendor, and what it means for telco operations.

THIRD-PARTY RISK

Who owns vendor risk?



One owner

One person in the business signs off each vendor.



Access list

Who can log in remotely, to what, and when.



On request

Switch vendor access on when needed, then off.



Remove on exit

When a contract ends, remove the login that day.

BEFORE YOU SIGN

Six questions to ask a vendor

Passwords?

Does each unit get its own password?

Encryption?

MQTT on port 8883 with TLS? HTTPS?

Updates?

Are they signed? For how many years?

Remote access?

Who on your side can log in, and how?

Our data?

Where is it kept? Can we take it with us?

Weaknesses?

How do we report one? How fast is the fix?

RULES THAT APPLY HERE

Security rules in Malaysia

Cyber Security Act

Act 854 (2024): critical sectors must assess risk and report incidents.

PDPA (2024)

Personal data, such as camera images: report a breach within 72 hours.

MCMC codes

Technical codes on IoT security, from MCMC.

A summary, not legal advice. Abroad: the UK PSTI Act and the EU Cyber Resilience Act.

TODAY → WITH IOT → THE BENEFIT

In telco operations

Today	With IoT, done safely	The benefit
Site access: a break-in is found at the next visit	A door contact alarms the NOC at once	Less theft, faster response
Vendor access: contractor logins stay open for years	One login per contractor, only for the job	Fewer open doors
Alarms: fuel loss is found late, or a reading is wrong	Encrypted, trusted fuel and battery alarms	Fewer wasted site visits

TEAM ACTIVITY · 10 MIN

Spot the vulnerability

1. Get into your **mixed teams of three**. Rotate roles.
2. Each team takes **2 of the 5 snippets**.
3. For each one: **what's wrong?** What's the **worst case**?
4. **How would you fix it?** Every fix is in this module.
5. Decision owner: **ready for tower sites**, or not yet?

DEBRIEF

Five snippets, five fixes

Wi-Fi on GitHub

Keep secrets in a separate file, never shared.

admin / admin

Force a new, unique password at setup.

Plain HTTP

Use HTTPS, and MQTT on 8883 with TLS.

One lock topic

Each lock gets its own identity and topic.

Unsigned update

Sign it, use HTTPS, check the signature.

All five are on the **OWASP IoT Top 10**, a free checklist of common IoT weaknesses.

BUSINESS-CASE CANVAS · 3 MIN

Canvas box 8: the risks

1. Take your team’s canvas and find **box 8: risks**.
2. **Security:** which door could an attacker use? Who has access?
3. **False alarms:** what if it alerts too often, or too late?
4. **Vendor lock-in:** can you change supplier and keep your data?
5. For each risk, write **one fix** and **one owner**.

KEY TAKEAWAYS

What to remember

- A cheap sensor at a remote site is a computer: it can be **the way in**
- Mirai: **default passwords** alone built a huge botnet
- Target: the way in was a **contractor’s login**
- **Authentication** proves who; **authorisation** limits what
- Keep **site systems (OT)** apart from office IT; **encrypt** MQTT
- **Vendor risk** needs an owner, and questions before you buy

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Name two security risks at a remote tower site.

Q2 Why did the Mirai botnet take over so many devices?

Q3 How did the Target breach begin? Who has a similar key to our sites?

Q4 Authentication or authorisation? Give a tower-site example of each.

UP NEXT · MODULE 8

Real-World IoT Applications and Project Showcase

Complete systems for tower sites and data centres, Lab 7, and your canvas pitch.

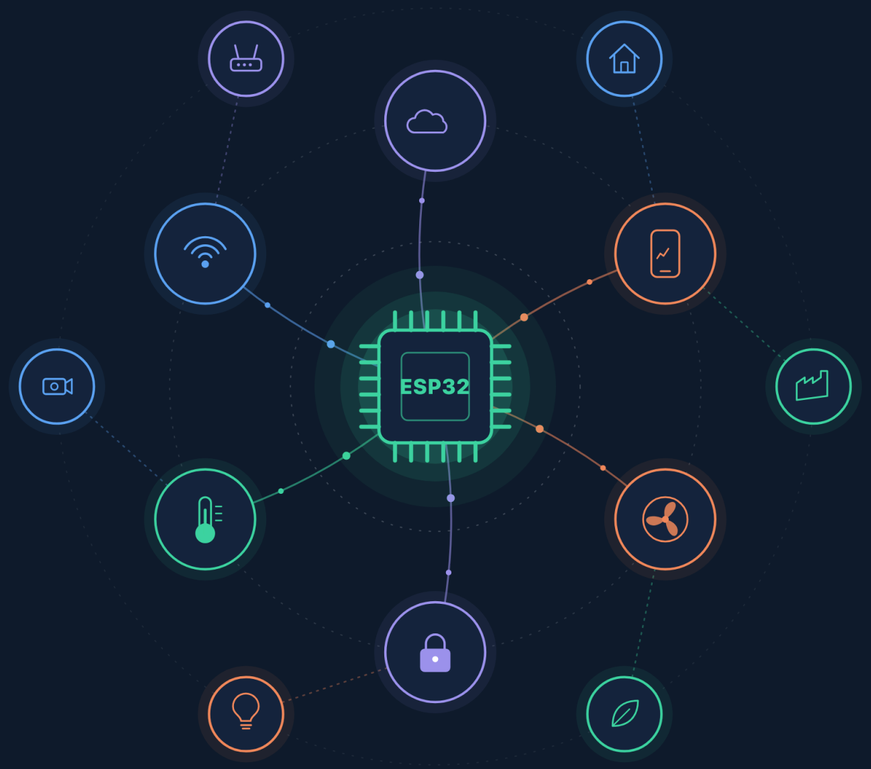
Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop

MODULE 8 · 45 MINUTES

Real-World IoT Applications and Project Showcase

Design a real IoT system, build the Telecom Site Monitor, and pitch your business case.



Mehdi Lotfinejad

Introduction to the Internet of Things

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Eight modules, one Smart Site Monitor

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Lab 5

07
Security
Group activity

08
Real world + pitch
You are here

BY THE END OF THIS MODULE

You will be able to...

01 **Use** a simple design method: problem, requirements, layers

02 **Explain** IoT choices for towers, assets and data centres

03 **Build** the Telecom Site Monitor in Lab 7

04 **Estimate** the costs and the payback of your idea

05 **Pitch** your business case in two minutes

Part A

A design method you can reuse

Problem → requirements → each layer, chosen for a reason.

THE DESIGN METHOD

Problem, requirements, then each layer

Problem

Who has it? What does it cost today?

Requirements

What must it do, where, how fast?

Layers

Device, link, edge, cloud, app.

The why

Why this, not that?

Your canvas follows the same order: box 1 is the problem, boxes 2–5 are the design.

STEP 1 · THE PROBLEM AND THE PEOPLE

“When the grid fails at night, nobody sees the battery run down until the tower site goes off air.”

Ask: what problem? Who has it? What does it cost?

Part B

Real sites, real decisions

The tower site, assets on the move, our own data centres — and a farm.



Photo: pslim, CC BY 2.0, via Wikimedia Commons

SECTOR CASE 1 · SMART TOWER MONITORING

The tower site

- Sense: battery, fuel level, door, shelter heat
- Tower **RMS** sends every site’s alarms to the **NOC**
- Energy: **15–20%** of operator costs (GSMA Intelligence, 2025)
- Fuel drops, generator off? **Possible theft**
- Battery trend, run-hours: **fix before it fails**

PAUSE AND CHECK

At a tower site, where should each rule run: on site, or in the NOC? (a) Start the generator on low battery. (b) Spot fuel theft.

Think about what happens when the link to the site is down. Thirty seconds with your team.

SECTOR CASE 2 · ASSET TRACKING

Where is it, and is it moving?



Generators

Mobile units sent where the grid fails



Batteries

Valuable, and a target for theft



Spare parts

Which store has the part we need?



Field vehicles

Which team is nearest the fault?

A small **cellular IoT tracker** (Module 4): its position, and an alarm if it moves when it shouldn’t.



Photo: Daoducquan, CC BY-SA 4.0, via Wikimedia Commons

SECTOR CASE 3 · THE TELCO'S OWN DATA CENTRES

Data centres and exchanges

- Sense: inlet air, humidity, leaks, **UPS** health
- Edge: a local alarm and extra cooling
- **DCIM** software: every rack on one screen
- Server inlet air: **18–27 °C** (ASHRAE)
- **PUE 1.56**: 0.56 kWh extra per 1 kWh of IT (Uptime, 2024)

Every choice traces back to a requirement

Layer	Choice	Why
Devices	Soil and temperature sensors; relay for the valve	Measures what matters; switches the valve safely
Connectivity	**LoRaWAN** to one farmhouse gateway	Kilometres of range, years on a battery
Edge	Open below **30%** moisture, close above **45%**	Works if the internet drops; no flickering
Cloud	Store readings; alert if a sensor is silent for 2 h	Checks across all fields, plus history
Application	Phone dashboard: one status tile per field	See what needs attention in seconds
Security	Own keys per sensor; MFA; signed updates	Hard to fake readings; can be patched



Photo: Christianjhart1, CC0, via Wikimedia Commons

THE SAME DESIGN AT PLANTATION SIZE

Smart agriculture

- Soil, weather and tank-level sensors
- **LoRaWAN** or **NB-IoT**: big farms, small messages
- Watering decided **locally**; cloud adds forecasts
- Low-power cellular IoT: **1 billion** connections by the end of 2025 (GSMA)

IN TELCO OPERATIONS

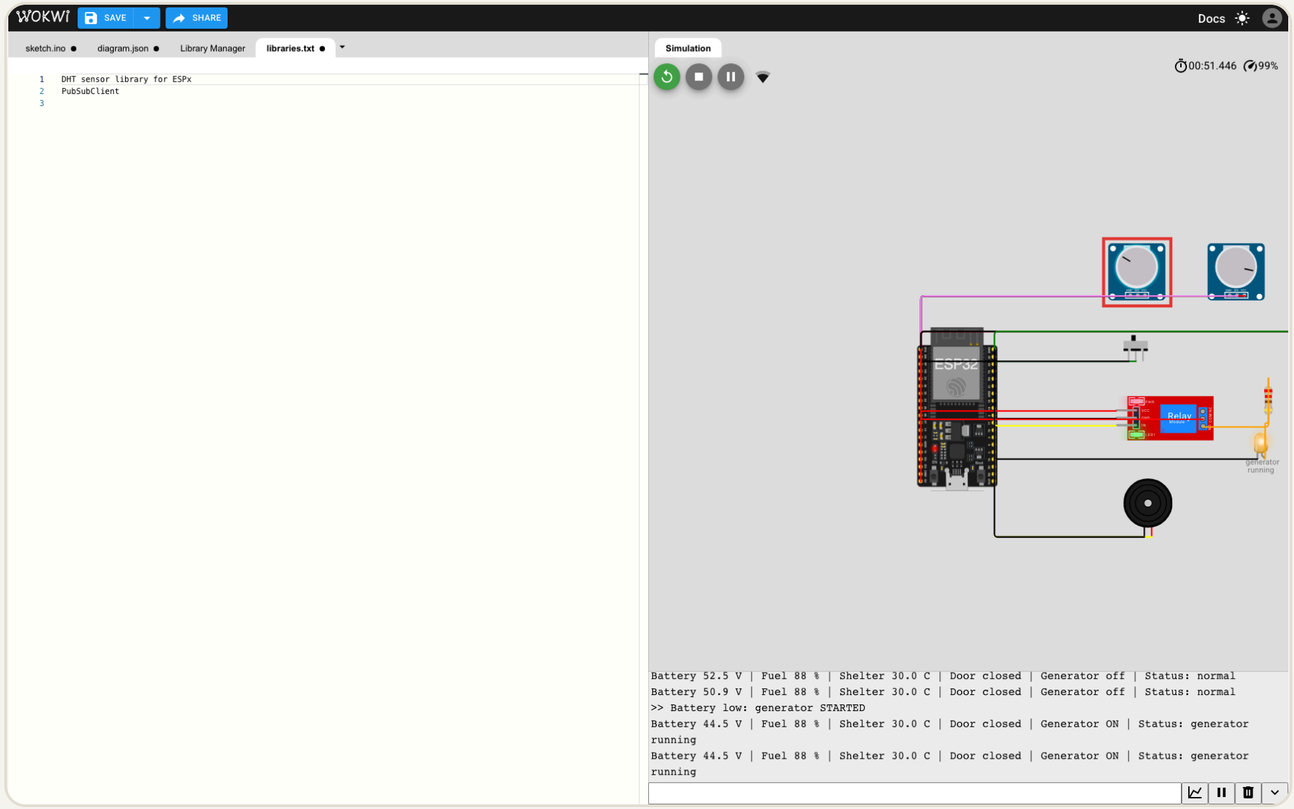
Today, with IoT, and the benefit

Operation	Today	With IoT	The benefit
Site energy and diesel	Fuel and batteries checked on visits	Live battery and fuel; generator only when needed	Less diesel, fewer site visits
Site security	Theft found at the next visit	Door and theft alarms reach the NOC	Less theft, faster action
Maintenance and spares	Repair after it fails; hunt for parts	Battery trends, run-hours, tracked spares	Faster repair, better use of engineers

No numbers yet: your team estimates them for its own case on the canvas.

LAB 7 · 18 MINUTES HANDS-ON

Capstone: the Telecom Site Monitor



1. New ESP32 project: paste sketch.ino and change **YOUR_NAME**
2. Paste diagram.json, add libraries.txt, press Play
3. Generator off: turn the **right dial** (fuel) down → theft alarm
4. Turn the **left dial** (battery) below **47 V** → the generator starts
5. Click the slide switch → door alarm; then watch it all in HiveMQ

Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

BUSINESS-CASE CANVAS · BOXES 6 AND 7

Costs and payback

Devices

Sensors, boards, installation

Connectivity

A fee per device, every month

Platform

Cloud, RMS or DCIM software

People

Set-up, checks, training

7. Payback

Total cost ÷ saving per month

Box 6 is every cost, for years — not only the device price. Round numbers are fine.

THE PITCH · 2 MINUTES PER TEAM

The committee listens for three things



Clear problem

Who has it, what it costs



One number

A saving you can explain



One named risk

From box 8, with your fix

The steering committee: your trainer and, ideally, a sponsor from the business.

BUSINESS-CASE CANVAS · 13 MIN

Finish your canvas, then pitch

1. Boxes 6 and 7: list the costs, estimate the payback
2. Choose a presenter; the team holds the canvas
3. Pitch to the steering committee: 2 minutes
4. Committee: problem? number? risk? One line each
5. Vote: which pilot would you fund first?

KEY TAKEAWAYS

What to remember

- Start with **the problem and its cost**, not the technology
- Choose each layer **for a reason** you can say aloud
- Tower, assets, data centre, farm: **same five layers**
- Site data has a price: **energy, fuel, outages, visits**
- A good pitch: **clear problem, one number, one risk**

CHECK YOUR UNDERSTANDING

Quick quiz

Q1 Which two readings can reveal fuel theft at a tower site?

Q2 Why should the generator start on site, not from the NOC?

Q3 A data centre has a PUE of 1.56. What does that mean?

Q4 What three things did the steering committee listen for?

THANK YOU

Thank you — and your next steps

Take home your canvas and the 2-page manager brief. The course materials — book, labs and slides — stay available.

Mehdi Lotfinejad

Book, labs and slides: github.com/datatweets/intro-to-iot-workshop